CHAPTER 4 WIRE ROPE 4-1

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Chapter 4 Wire rope

Summary：

The wire rope is of consumable part and is not in the scope of warranty.

The brand of wire rope is always designated by the user according to their own experiences that may various from terminals to terminals. And the user usually purchases new wire ropes for replacement. As one of the most important parts for this quayside container crane, ZPMC pays sufficent attention to it. This is also the main reason why the wire rope is instructed in an independent chapter.

In this chapter, the basic knowledge of handling, transportation, maintenance, inspection, changing of wire rope will be introduced. And the **inspection, lubrication, adjustment and re-reeving of wire rope will be focused on in this chapter**.

	**This chapter will introduce the common knowledge of the wire rope. If the conflict exists, the instructions provided by the wire rope supplier shall be preferential. There may be different regulations for the usage, maintenance, discarding of the wire rope that shall be strictly followed in different countries.**

4.1 Basic knowledge of wire rope

4.1.1 Construction

The constriction of an example is shown in Figure 4-1.

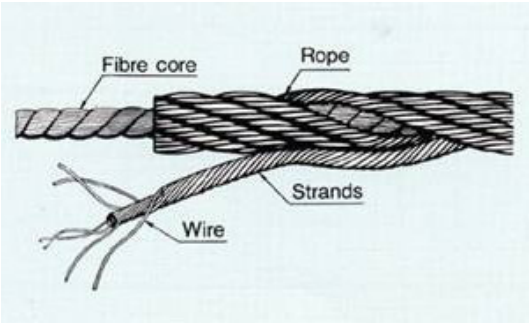
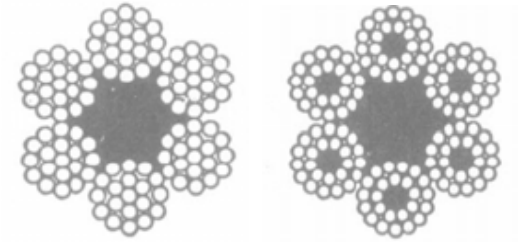


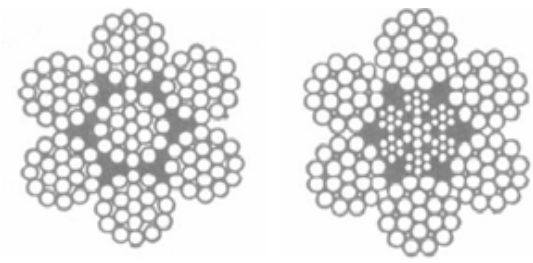
Figure 4-1：Wire rope appearance (example)

4.1.2 Core

The core of the wire rope used for quayside container crane are shown in Figure 4-2（d)。



(a) Fiber core (b)Wire and Strands fiber core



(c) Independent Wire Strand Cores(IWSC) (d) Independent Wire Rope Cores(IWRC)

Figure 4-2：Core of wire rope

4.1.3 Type of laying

1) Direction of Laying

As concerns the direction of laying the rope and the strand, there is the Z lay and the S lay, as shown in Figure 4-3. in the case otherwise specified, the rope shall be made of Z lay and the stand products of the S lay.

2) Laying of rope

There are two types of lay as shown in Figure 4-3.

- Ordinary lay(or regular lay)：The lay of the rope and the lay of the strands are in opposite directions.
- Lang's lay：The lay of rope and the lay of the strands are in the same directions.



a) Ordinary Z lay/ordinary S lay b) Lang's Z lay/ Lang's S lay

Figure 4-3：Laying of wire rope

3) The length of a laying

Generally, The length of a laying (length of rope pitch) is usually 6.2~8.2 times of the wire rope diameter. The measurement of the length of a laying is shown in Figure 4-4（For a wire with 6 Srtands)：

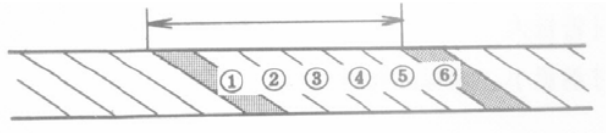


Figure 4-4：Measurement of a laying

Table 4-1：Comparison of the characterizes by laying of rope

Item	Ordinary lay	Lang's lay
Appearance	The wires are approximately parallel to the axis of the rope	The wires form a certain aggle with the rope axis
Merits	Kinks do not form easily and it easy to handle. The lays are tight and the form does not crumble.	The wires appearing on the surface are long and superior in wear resistance. Flexible and has good fatigue resistance nature.
Demerits	The wires appearing on the surface are long and superior in wear resistance. Flexible and has good fatigue resistance nature	The rope has a large torque and forms kinks easily

4.2 Handling of wire rope

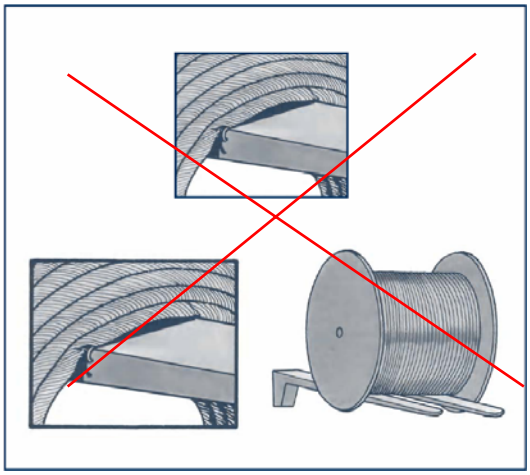
4.2.1 Transportation and storage

To prolong their life, the wire rope must be properly transported, stored, cleaned, inspected and installed. Wire ropes are easily affected by external damage. Therefore, they must be handled with utmost care during transport and unloading. All wire ropes should be stored clean, dry and cool and soil contact must be avoided anyway. Wire ropes must always be handled with equipment that will not damage the rope.

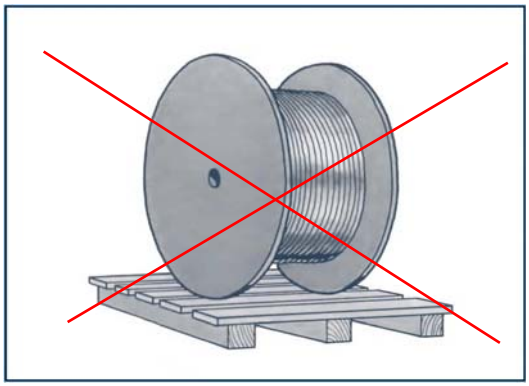
Reels of wire rope should be stored in an area where they will not be damaged by objects falling, chemical spills, excessive heat, open flames, welding, and excessive heat. Reels should be stored vertically on their flanges and covered to prevent dirt and debris from embedding into the wire. When storing reels, ensure that one reel will not roll into another. This will prevent one reel flange from hitting the wire rope surface of another reel. If necessary, reel flanges should be choked to prevent movement. The wrong method as shown in Figure 4-5 shall be prohibited. For the correct method handing the rope, please refer to Figure 4-6.

Do not store wire rope in atmosphere of volatile chemical or moisture. Wire rope will rust.

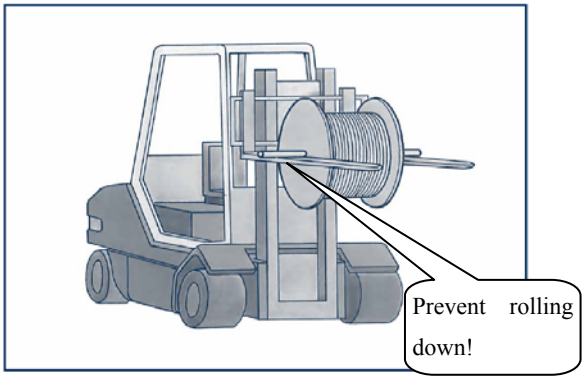
Do not use any detergent oil to clean wire rope. Detergent oil soaks into and dissolves the lubricant inside the rope and will cause internal corrosion. Clean with a wire brush.



(a)

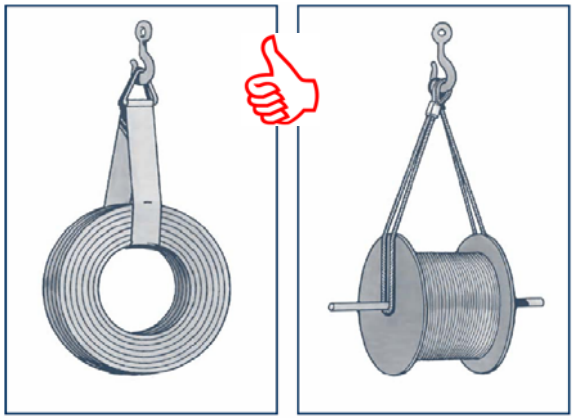


(b)



(c)

Figure 4-5：Un-recommend method



(a)



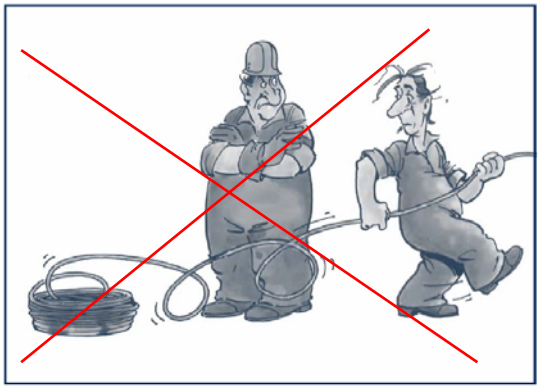
(b)

Figure 4-6：Recommend method

4.2.2 Installation of wire rope

Proper/improper methods of unwinding a wire rope from a reel are separately introduced in Figure 4-7, Figure 4-8.. Considerable care must be taken when unwinding a wire rope. Their operational performance is substantially influenced by the way they are handled. Bending or twisting wire rope can cause internal damage that will adversely affect the operational life of the wire rope. Avoid unwinding wire rope on the ground to protect the wire rope from debris. If there is no other option, make sure the ground is clean or covered with boards or plastic sheets.

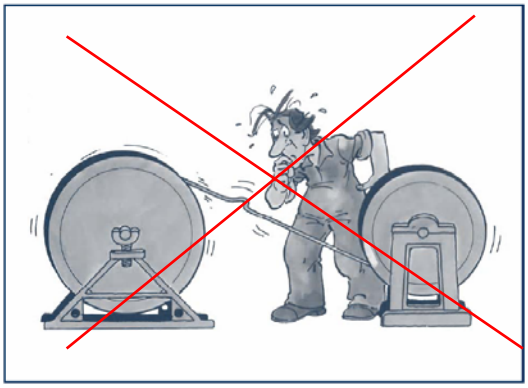
Every wire rope bends in a particular direction based on the lay of the rope. So, the wire rope should always be uncoiled that coincides with their lay. As shown in Figure 4-8, the prohibited methods of unwinding a wire rope, wire rope should never be pulled out with the reel fixed.



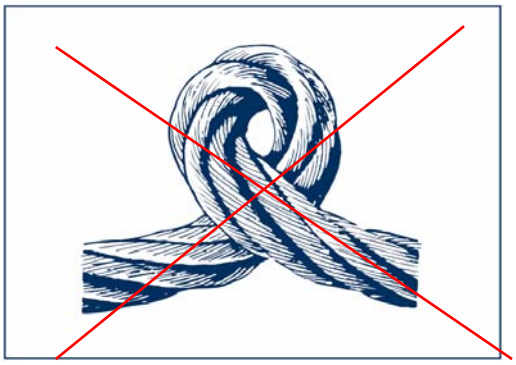
（a）



（b）



（c）



（d）

Figure 4-7：Improper method



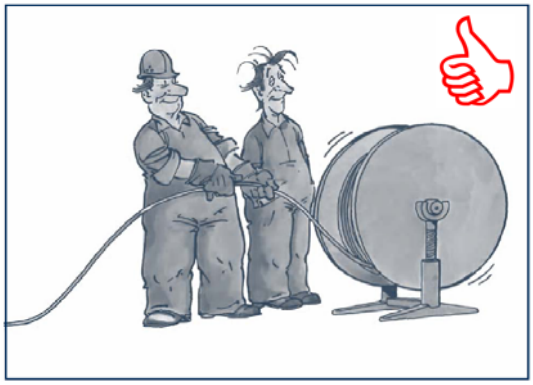
（a）



（b）



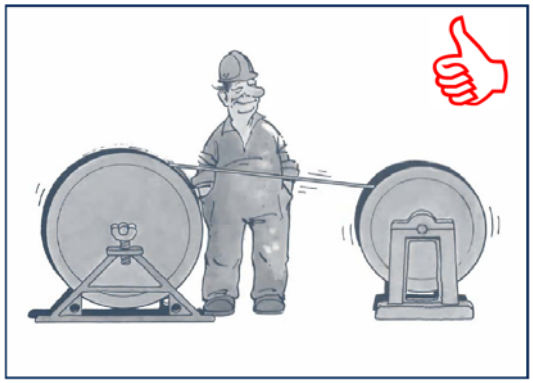
（c）



（d）



（e）



（f）

Figure 4-8：Proper method

If wire rope is unreeled with the reel fixed, wire rope will be tangled and will cause kink or deformation.

4.2.3 Termination and treatment of wire rope end

4.2.3.1 The wire rope clip

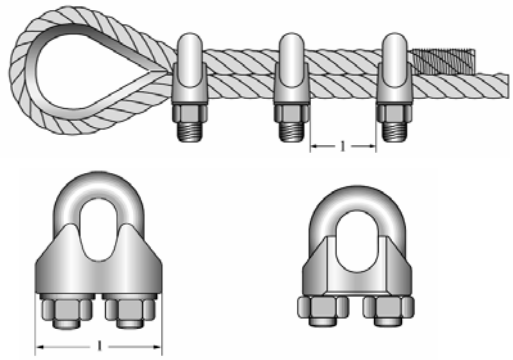


Figure 4-9：Distance between clips

Rope end connections using wire rope clips are very popular because they can be fabricated on-site with very little effort and they are also very cheap. Compared with many other end connections they are easily detachable and can be inspected without difficulties. The rope end is always laid around a thimble and the first clip is threaded on close to the thimble. The distance between clips is approximately equal to the width of the clip as shown in Figure 4-9. The minimal quantity of clips used for an end and the minimal tighten torque shall follow the values as shown in Table 4-2.. The incorrect installation of clips such as shown in Figure 4-10 may reduce its breaking load considerably and results in unexpected accident.

Table 4-2：Tightening torques for rope clips

Nominal size	Torque [Nm]	clips number
5	2.0	3
6.5	3.5	3
8	6.0	4
10	9.0	4
13	33	4
16	49	4
19	67.7	4
22	107	5
26	147	5
30	212	6
34	296	6
40	363	6

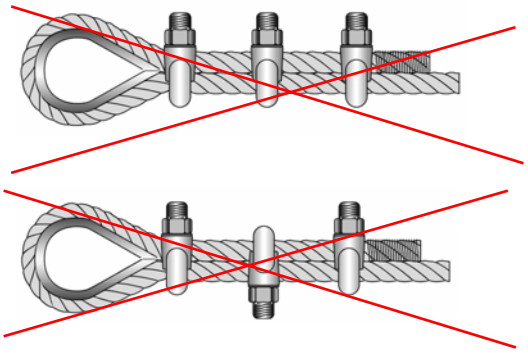


Figure 4-10：End connections with rope clips fitted incorrectly

4.2.3.2 The asymmetrical wedge socket

When terminating a rope end with a wedge socket, ensure that the rope tail can not withdraw through the socket by securing a clamp to the tail or by following the manufacturer's instructions. The tail length should be a minimum of 20 rope diameters for all Rotation Resistant wire rope and a minimum of 6 rope diameters for 6 and 8 strand ropes.

See Figure 4-10 for two recommended methods of securing the rope tail of a wedge socket termination.

The installation as shown in Figure 4-12 shall be prohibited in any case.

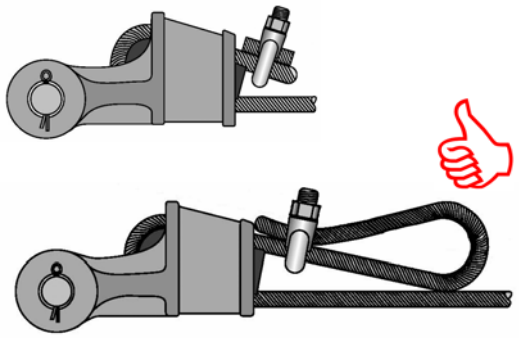


Figure 4-11：Recommended clipping of the 'dead' line

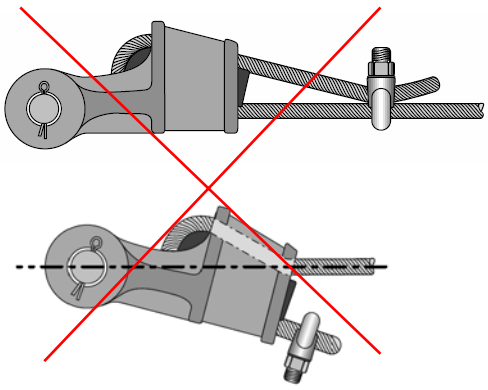
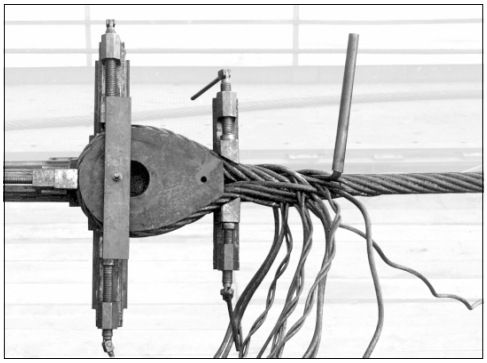


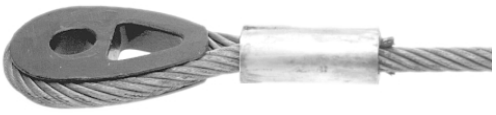
Figure 4-12：Incorrect installation of a wedge socket connection

4.2.3.3 Other methods of rope end connection

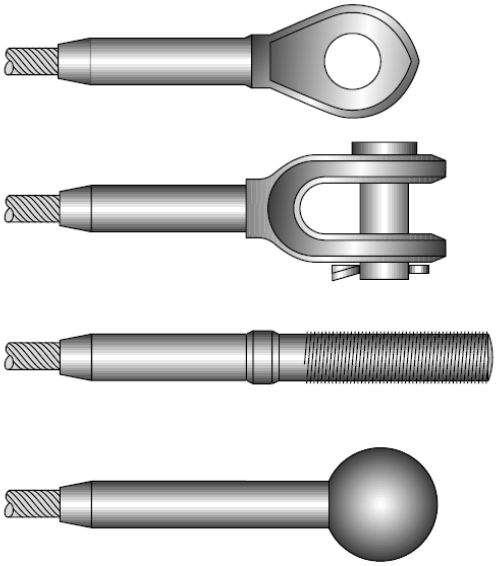
There are also many other methods for the rope end connection as shown in Figure 4-13 such as:（a）The splice，the oldest end connection for ropes, is not used very often and may only limited to some wire ropes in very small size and used for secondary application in the container cranes. (b) The aluminum clamp that usually shall be fabricated by the specially machine.(c) swaged sockets and etc.



(a)



(b)



(c)

Figure 4-13：Other rope end connections

4.2.4 Wire rope cutting and treatment of rope end

In every case, each side of the cut must be properly seized to prevent strand disturbance before cutting. After all the seizing are secure, then you may cut the rope. Normally, one seizing on each side of the cut is sufficient. For non-preformed or rotation-resistant ropes, a minimum of two seizing on each side is recommended. These should be spaced six rope diameters apart. The procedures of cutting are below:

1) The rope and this wire end are now wrapped moving away from the location of the intended cut. Keeping wraps parallel, close together and in tension. Twist ends of strand together by hand.
2) The rope is tightly wrapped for a distance of approx two three rope diameters
3) Both ends of the seizing wire are then pulled tight and twisted together. The twisted connection is then hammered into a strand valley.
4) After preparing the other side of the intended cut accordingly the rope can now be cut. Please also see Figure 4-14 for details

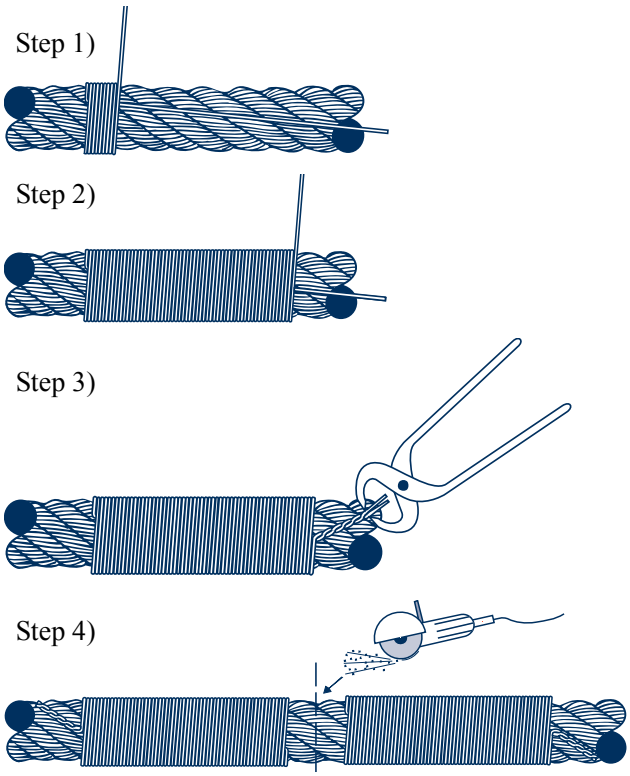


Figure 4-14：Wire rope cutting

Most ropes are shipped with the ends seized as they are prepared for cutting as shown in Figure 4-15（a）. You can usually install seized ropes without further preparation. In some cases, though tight openings in drums and wedge sockets- or even complicated reeving systems – require special end preparation. Then, the strands must be tightly held without increasing the rope diameter. In such cases, the ends are tapered and welded as shown in Figure 4-15 (b) (c), or the ends fused as shown in Figure 4-15(d). It's sometimes necessary to provide a loop or link to which a lighter line is fastened to pull the rope into place or around sheaves. Some of these special end preparations are shown as follows:



(a) Seized end



(b) Tapered and welded end



(c) Seized and torch cut with ends fused



(d) Steel end link

Figure 4-15：Treatment of rope end

4.3 Inspection of wire rope

A steel wire rope is a commodity with a limited lifespan. There are many factors that may influence the rope life and performance such as: the quality of the rope, load during the operating, operational environment, maintenance etc. Many properties will change and many damages may have during its service especially before the failure of the rope. Those changes and damage may be the premonition of the failure of a wire rope. So regular inspection is very necessary to eliminate any hidden defect to extent the rope life or find premonition of rope failure before rope changing. The factors influence the rope life, the properties changes and the damages are also recorded during the inspection so the reasons of rope failure can be analyzed and the deterring factor can be reduced to the minimal.

	**The properties of wire rope will rapidly decrease before reaching the rope life time. So the inspection intervals need to be shortened before rope replacement for the safety.**

4.3.1 The factors influence the rope life

There are many factors that may influence the rope life and performance including the condition of operating, maintenance, working environment etc. and is very complicate. To a rope on shore container crane, some main factors: are below:

The ropes are subjected to the changing load when running through drums and sheaves continuouslyso the

ropes are bended constantly during all the operation and is easily fatigue.

1) The corrosion due to the humidity and salt spray in a marine environment: The air is mixed with humidity and salt which is liquefiable in water. The ions in water will speed up the diffusion of oxygen ion and that also speed up the oxidation reaction of wire rope and worse the corrosion of it.

2) Affect of foreign objects embed in the wire rope. The wire ropes are working on an opening adverse polluted environment. The wind are also very stronger that may bring the hard foreign objects such as graves, slag and etc to the surface of the wire rope and then go to the sheaves, drums, or even inside of the wire ropes that may cause wear and wire broken.

3) Affect of frequency operating: There is only a thin film of lubricant on the surface of the wire rope. The lubricant is squeezed out from internal of rope when the rope is tensioned and drawn in when the rope is loosed. This is frequency duplicated and not only causes the escape of the lubricant inside the rope but also speeds up the oxidation of the lubricant.

Of course, there are many factors that may affect the life of the wire rope. Those factors may not only related to the design but also including a lot of abnormal factors such as:

1) **Multi-layers** of the rope on the drum can result in severe distortion in the underlying layers.

2) **Bad spooling** (due to excessive fleet angles or slack winding) can results in mechanical damage, shown as severe crushing, and may cause shock loading during operation.

3) **Small diameter sheaves** can result in permanent set of the rope, and will certainly lead to early wire breaks

4) **Oversize grooves** offer insufficient support to the rope leading to increased localized pressure, flattening of the rope and premature wire fractures. Grooves are deemed to be oversize when the groove diameter exceeds the nominal rope diameter by more than 15%.

5) **Undersize grooves** in sheaves will crush and deform the rope, often leading to two clear patterns of wear and associated wore breaks.

6) **Excessive fleet angle** can result in severe wear of the rope due to scrubbing against adjacent laps on the drum. Rope deterioration at the termination may be exhibited in the form of broken wires. An excessive angle of fleet can also induce rotation causing torsion imbalance.

7) Using the wire rope of incorrect size, construction, or grad .

8) Dragging rope over obstacles.

9) Having improper lubrication

10) Operating over sheaves and drums with improperly fitted grooves or broken flanges.

11) Jumping off sheave

12) Subjecting it to acid fumes.

13) Attaching fittings or installation improperly.

14) Subjecting it to severe or continuing overload .

4.3.2 Inspection Method

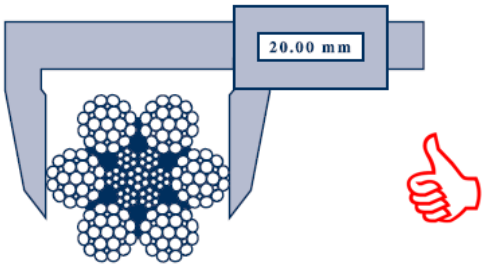
The measurements shall be taken by the qualified people.

1) Measurement of diameter

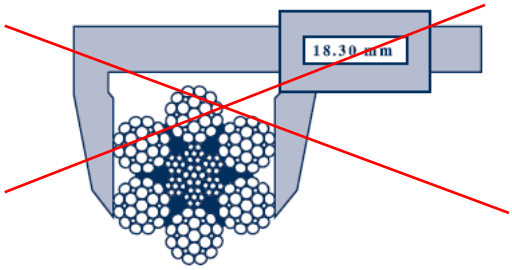
The size of a wire rope is designated by its diameter. The diameter of a wire rope should be measured by a set of Vernier calipers.

The correct measurement of the rope diameter is shown in Figure 4-16 (a). The measurements as shown in Figure 4-16 (b) are wrong.

☞	**The nominal diameter of a new wire rope must be measured and recorded before it is installed on the crane. The results of every subsequent check should also be recorded in the journal for that wire rope.**



（a）Correct measurement of the rope diameter



（b）Wrong measurement of the rope diameter

Figure 4-16：Wire rope measurements

2) Visual inspection

Visual inspection is the method most wildly used. The wire rope surface shall be cleaned before inspection. Wire breaks of outer wire which do not occur at the crowns of the strands but at the contact points of two neighboring wires or even on the underside of the strands are very difficult to detect. With thin ropes, which can be completely unloaded, such wire breaks can be made visible by extreme bending of the rope as shown in Figure 4-17. The interior of wire rope can be examined according to Figure 4-18.

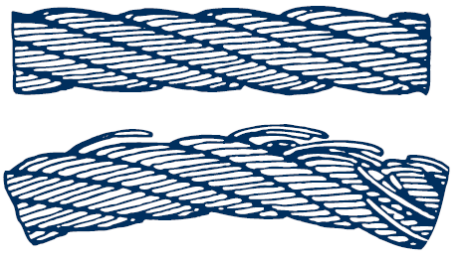
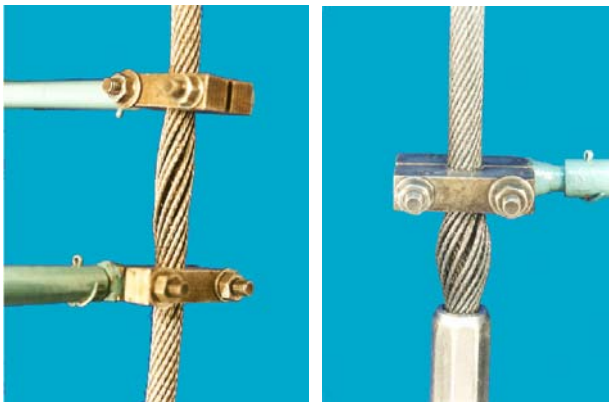


Figure 4-17：Detecting wire breaks by bending the rope



(a) (b)

Figure 4-18：Examine the interior of the wire rope

3) Tactile inspection

4) The Tactile inspection is often used to detect the wire breaks by spooling the rope through your hand (only if the rope can be run slowly enough). This method is usually to be used together with the visual inspection simultaneously. When applying this method extreme care is necessary as protruding wires from the rope construction can cause severe injuries. In many cases the inspector will run a piece of wood along the rope which is ten kicked back by the protruding wire break ends. Another useful technique is to hold a ring of cotton waste around the slow moving rope-any protruding broken wires will snag the cotton and thereby highlight the damaged area.

5) Magnetic testing of wire ropes（NDT）

There are some wire rope NDT machines to detect both surface and interior damages in the marketing. The

advantages of this instruments are they good at detect the damages not visible or interior the wire rope and even can inform the inspector about the extent of the rope deterioration. But the reliability of this kind of new techniques needs to be verified by the practice. For the details please reference the manuals of this kind of machine.



NDT using electromagnetic means may be used to detect broken wires and loss in metallic area. This method complements the visual examination but does not replace it.

4.3.3 Where and when must wire ropes be inspected?

Generally speaking the whole wire rope length must be visually inspected. It goes without saying that critical zones require extra attention. For example, critical zones are: Those rope zones which are exposed to the highest number of cycles, rope end connections, loading points; rope zones on compensation sheaves; rope zone on rope drums and etc. The rope zones on compensation sheaves are easily to be ignored because it is usually wrongly though “this zone are static and compensation sheaves do not work (as everybody known’s) permit a smaller diameter for this very sheave.”. So, the piece of rope in this zone not only carries out more bending cycles (especially due to the jumping) than the others but also is on a sheave with a smaller diameter. Special care shall also be taken to the rope zones in the grooves of sheave and drums nearby rope out points.

As far as possible, all visible parts of any rope shall be observed each working day with the objective of detecting general deterioration and deformation. Particular attention shall be paid to the rope at points of attachment to the crane.

The wire rope shall be inspected one week after fisting using or the wire breaks firstly be found.

After first inspection, inspect the wire rope monthly.

After abnormal loading or whenever an accident such as snag or in the case of presumed no-visible damage or installation of a new rope, the wire rope shall be inspected immediately. The intervals shall also be reduced after that.

The intervals shall also be shorted whenever any damage has occurred in connections with the reeving system especially before the period of the rope to be discarded.



The intervals of inspection can be adjusted according to the previous inspection record.

4.3.4 What shall be inspected?

A thorough inspection should cover the following points:

To detect the abrasion and breaks early and on time, a schedule shall be set for each rope. A thorough inspection should cover the following points:

1) Determining the rope diameter: Measure the diameter of wire rope and check the reduction in rope diameter and the rate of the reduction.

2) Is there the rope corrosion? Is the rope properly lubricated?

3) Abrasion includes outer and interior of the rope.

4) Are there any visible break wires? What kind of break wire it is? The rope shall be discarded if the break wire reaches a specified numbers.

5) If there is a phenomenon that the wire rope has been crushed, abraded, flattened, Basket or lantern deformed kinked or has sustained any other damage that distorts it; the rope should be discard.

6) Inspection the rope connections and parts connected with the wire: What’s condition of the connection? Damaged connection shall be changed or corrected immediately. Is the abrasion of the drum and sheave grooves reach

allowable tolerance?

All inspection results shall be well recorded in the daily log and special record for wire ropes

	**The results of every visual inspection shall be recorded for tracking purposes.**

A Typical example of rope examination record is given in ISO 4309:2004(E) Annex B that could be referenced. The wire rope discard date ad discard number of cycles could be predicted the series of wire rope inspection record. The records of inspection are also important material for analyzing the key factors affecting the rope life so the measures to improve the rope life can be found.

4.3.5 Discard criteria

The safe use of wire rope is qualified by the following criteria:

1) Broken wires: the nature and number of broken wires; broken wires at the termination; localized grouping of wire breaks; the rate of increase of wire breaks.

2) The fracture of strands;

3) Reduction of rope diameter, including that resulting from core deterioration.

4) Decreased elasticity.

5) External and internal wear;

6) External and internal corrosion.

7) Deformation: waviness; Basket or lantern deformation, core or strand protrusion /distortion, Wire protrusion; local increase in diameter of rope, flattened portions, Kinks or tightened loops; bends.

8) Damage due to heat or electric arcing

9) Rate of increase of permanent elongation.

All examinations shall take into account these individual factors, recognizing the particular criteria. However, deterioration frequently results from a combination of factors, giving a cumulative effect which should be recognized by the competent person, and which reflects the decision to discard the rope or to allow it to remain in service. Particular attention shall be paid to the abnormities that may damage the rope and shall be eliminated before rope changing.

The discard criteria can be found in ISO 4309:2004（E），DIN15020, DIN3088 and etc. The local criteria shall be taken priority of those criteria.

The content of discard criteria of ISO 4309:2004（E）is as below

4.3.5.1 Nature and number of broken wires

In the case of 6- and 8-strand ropes, broken wires usually occur at the external surface. In the case of rotation-resistant ropes, there is a probability that the majority of broken wires will occur internally and are “non-visible” fractures. Detail criteria are given in Table 4-3.

4.3.5.2 Broken wires at termination

If there are broken wires at, or adjacent to the termination, the cause of this deterioration shall be investigated and, where possible, the termination shall be remade, shortening the rope if sufficient length remains for further use, otherwise the rope shall be discarded.

4.3.5.3 Localized grouping of broken wires

Where broken wires are very close together, constituting a localized grouping of such breaks, the rope shall be discarded. If the grouping of such breaks occurs in a length less than 6*d* or is concentrated in any one strand, it may be necessary to discard the rope even if the number of wire breaks is smaller than the maximum number shown in Table 4-3. Pictures 3 in Table 4-4 show the typical localized grouping of broken wires. They are usually the main damages to the main hoist and trolley towing ropes.

4.3.5.4 Rate of increase of broken wires

In applications where the predominant cause of rope deterioration is fatigue, broken wires will appear after a certain period of use, and the number of breaks will progressively increase over time. In these cases, it is recommended that careful periodic examination and recording of the number of broken wires be undertaken, with a view to establishing the rate of increase in the number of breaks. This premise may be used to propose a future date for rope discard.

4.3.5.5 Fracture of strands

If a complete strand fracture occurs, the rope shall be immediately discarded.

4.3.5.6 Reduction of rope diameter resulting from core deterioration

Reduction of rope diameter resulting from deterioration of the core can be caused by:

1) Internal wear and wire indentation.
2) Internal wear caused by friction between individual strands and wires in the rope, particularly when it is subject to bending.
3) Deterioration of a fiber core;
4) Fracture of a steel core.
5) Fracture of internal layers in an otation-resistant rope.

If these factors cause the actual rope diameter to decrease by 3 % of the nominal rope diameter for rotation resistant ropes, or by 10 % for other ropes, the rope shall be discarded even if no broken wires are visible.

4.3.5.7 Decreased elasticity

Under certain circumstances elasticity decreasing is usually associated with the working environment, a rope can sustain a substantial decrease in elasticity and is thus unsafe for further use.

Decreased elasticity is difficult to detect. However, it is usually associated with the following:

1) reduction in rope diameter
2) Elongation of the rope lay length
3) Lack of clearance between individual wires and between strands, caused by the compression of the component parts against each other appearance of fine, brown powder between or within the strands
4) Increased stiffness

If any above phenomenon are observed or examiner has any doubt, advice shall be obtained from a specialist. Increased stiffness of a rope can lead to abrupt failure under dynamic loading and is sufficient justification for immediate discard.

4.3.5.8 External wear

See picture 13 in Table 4-4 for the reference. Wear is promoted by lack of lubrication, or incorrect lubrication, and also by the presence of dust and grit. Wear reduces the strength of ropes by reducing the cross-sectional area of the steel strands. If, due to external wear, the actual rope diameter has decreased by 7 % or more of the nominal rope diameter, the rope shall be discarded even if no wire breaks are visible.

4.3.5.9 External and internal corrosion

Corrosion occurs particularly in marine and polluted industrial atmospheres. It will diminish the breaking strength of the rope by reducing the metallic cross-sectional area, and it will accelerate fatigue by causing surface irregularities which lead to stress cracking. Severe corrosion can cause decreased elasticity of the rope.

Corrosion of the outer wires can often be detected visually. See picture 14 in Table 4-4.

See Picture 14 in Table 4-4 for the internal corrosion. It is more difficult to detect than the external corrosion which frequently accompanies it, but the following indications can be recognized:

1) Variation in rope diameter; in locations where the rope bends around sheaves, a reduction in

diameter usually occurs. However, in stationary ropes it is not uncommon for an increase in diameter to occur due to the build-up of rust under the outer layer of strands

2) Loss of clearance between the strands in the outer layer of the rope, frequently combined with wire breaks between or within the strands.

If there is any indication of internal corrosion, the rope should be subjected to internal examination as indicated in Figure. Confirmation of severe internal corrosion is justification for immediate rope discard.

4.3.5.10 Waviness

See picture 11 in Table 4-4 for reference. In the case of waviness (see Figure 4-19), the wire rope shall be discarded if, under any load condition on a straight portion of rope that does not bend around a sheave or drum, the following condition is met: d1>4d/3 or d1>1.1d.

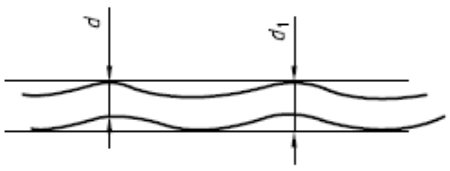


Figure 4-19：Waviness

4.3.5.11 Basket or lantern deformation

See picture 12 in Table 4-4. Basket or lantern deformation, also called “birdcage”, is a result of a difference in length between the rope core and the outer layer of strands. Different mechanisms such as excessive fleet angle that cause the rope touch the flange or the rope running over a “tight sheave”. The rope shall be discarded if the basket or lantern deformation occurs.

4.3.5.12 Core or strand protrusion/distortion

See picture 5, 7 in Table 4-4. Rope with core or strand protrusion/distortion shall be immediately discarded.

4.3.5.13 Wire protrusion

See picture 4 in Table 4-4. In wire protrusion, certain wires or groups of wires rise up, on the side of the rope opposite to the sheave groove, in the form of loops. Rope with wire protrusion shall be immediately discarded.

4.3.5.14 Local increase in diameter of rope

See picture 17 in Table 4-4. This condition usually relates to a deformation of the core such as in particular environments, a fiber core can swell up owing to the effect of moisture. If this condition causes the actual rope diameter to increase by 5 % or more, the rope shall be immediately discarded.

4.3.5.15 Flattened portions

See picture 17 in Table 4-4. Flattened portions of rope which pass through a sheave will quickly deteriorate and may damage the sheave. In such cases the rope shall be discarded immediately. Flattened portions of rope in standing rigging can be exposed to accelerated corrosion, and shall be subject to inspection at a prescribed shortened frequency if retained in service.

4.3.5.16 Kinks or tightened loops

See picture 18 in Table 4-4. Kink or tightened will cause imbalance of lay length, which will cause excessive wear, and in severe cases the rope will be so distorted that it will have only a small proportion of its strength remaining. Rope with a kink or tightened loop shall be immediately discarded.

4.3.5.17 Bending

Bends are angular deformations of the rope caused by external influence. Rope with a severe bend will suffer similarly to flattened portions of rope and should be treated as in clause 4.3.5.15.

4.3.5.18 Damage due to heat or electric arcing

Ropes that have been subjected to exceptional thermal effects, externally recognized by the colours

produced in the rope, shall be immediately discarded.

Table 4-3: Number, reached or exceeded, of visible broken wires in single-layer and parallel-closed ropes working in steel sheaves, signaling mandatory discard of rope[1]

Rope category Number RCN[2]	Number of load –bearing wires in all outer strands of the rope[3] (n)	Number of visible broken wires[4]			
		Mechanism group of M1~M4		Mechanism group of M5~M6	
		in rope over a length of [5]		in rope over a length of	
		6d	30d	6d	30d
01	n ≤ 50	2	4	4	8
02	51≤n ≤ 75	3	6	6	12
03	76 ≤ n ≤ 100	4	8	8	8
04	101 ≤ n ≤ 120	5	10	10	20
05	121 ≤ n ≤ 140	6	11	11	22
06	141 ≤ n ≤ 160	6	13	13	26
07	161 ≤ n ≤ 180	7	14	14	28
08	181 ≤ n ≤ 200	8	16	16	32
09	201 ≤ n ≤ 220	9	18	18	36
10	221 ≤ n ≤ 240	10	19	19	38
11	241 ≤ n ≤ 260	10	21	21	42
12	261 ≤ n ≤ 280	11	22	22	44
13	281 ≤ n ≤ 300	12	24	24	48
21	4-strand ropes，n ≤100	2	4	4	8
22	3 or 4- strand ropes，n ⩾100	2	4	4	8
23	At least 10 outer strands	2	4	4	8

1: Above values may also apply to ropes working in sheaves that are exclusively made of a polymer in combination with multi-layer coiling. They do NOT apply to ropes working in sheaves that are exclusively made of a polymer material in combination with single-layer coiling

2. See ISO 4309:2004(E) Annex E for detail rope category number RCN. Where:

09-Including rope of 6X36WS-IWRC(line contract) / 6xK36WS-IWRC(compacted strands) usually used for main hoist, trolley and boom hoist.

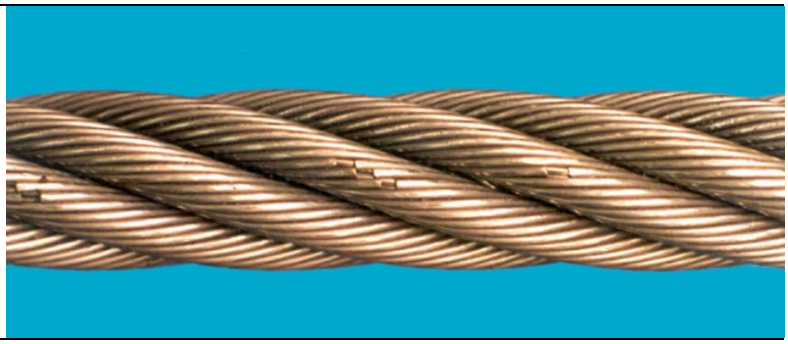
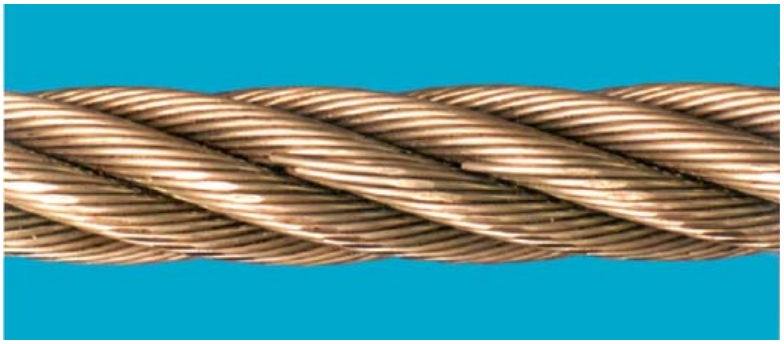
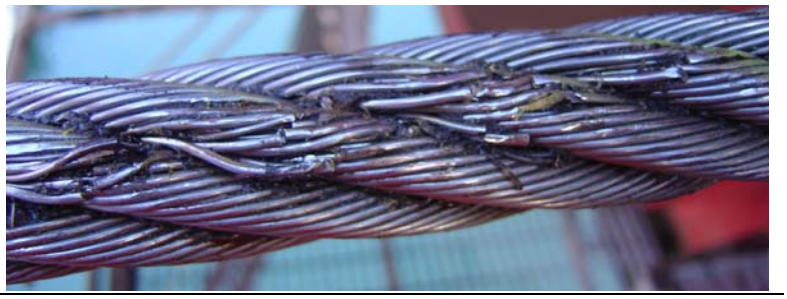
21~23-Rotation-resistant

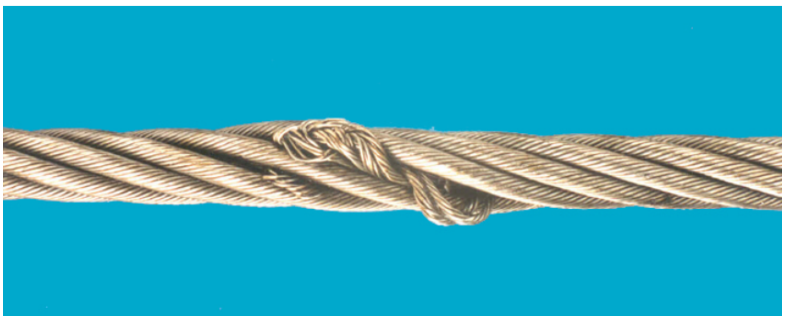
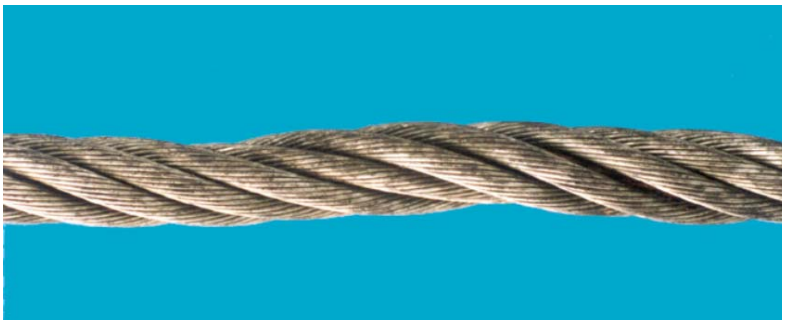
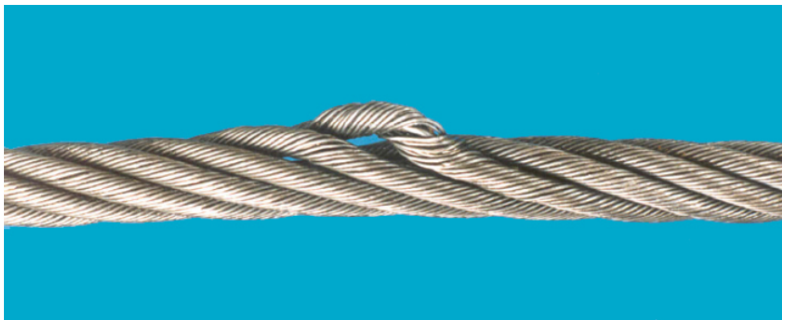
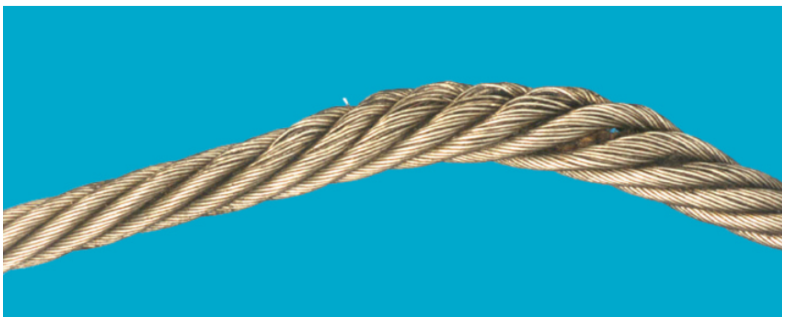
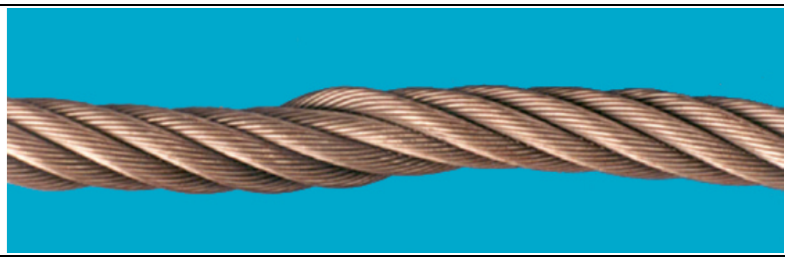
3: Filler wires are not regarded as load-bearing wires and are not included in the value of n. In ropes having a number of layers of strands, only the visible outer layer is considered in the value of n. In ropes having a steel core, this is regarded as an internal strand and is not considered in the value of n.

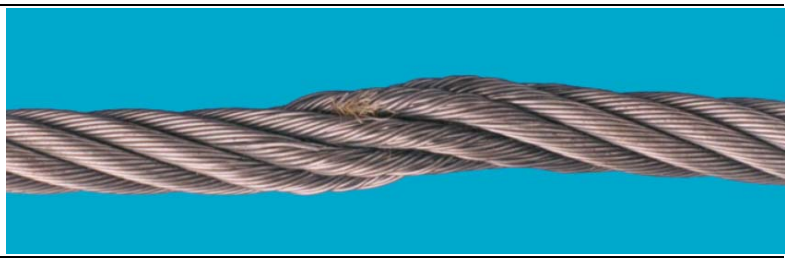
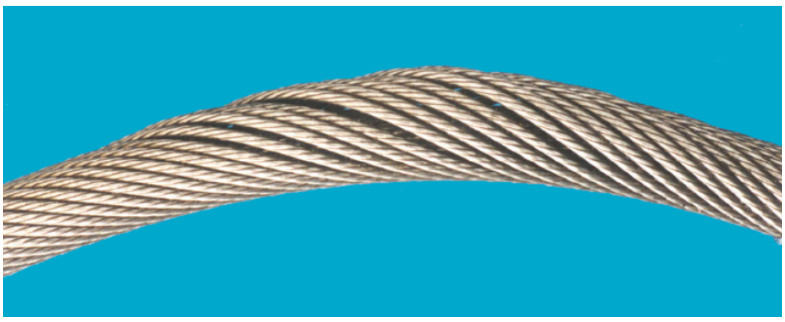
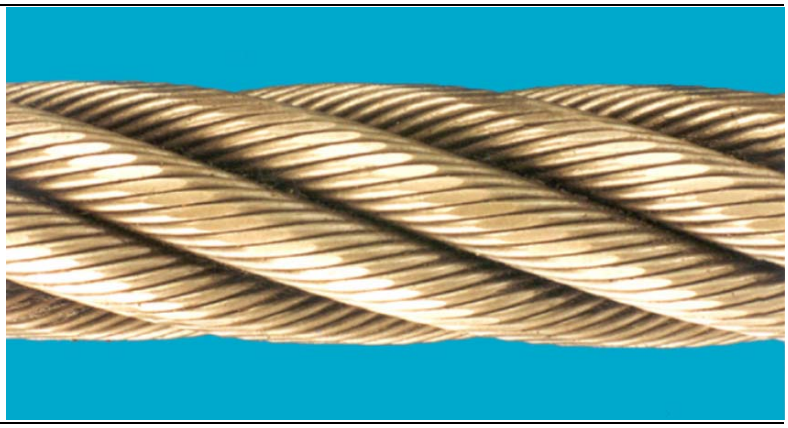
4: A broken wire will have two ends.

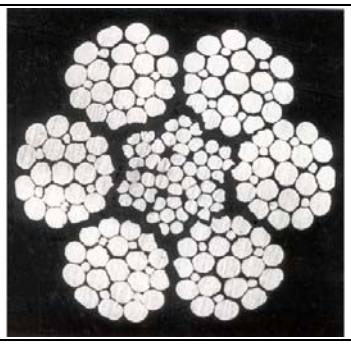
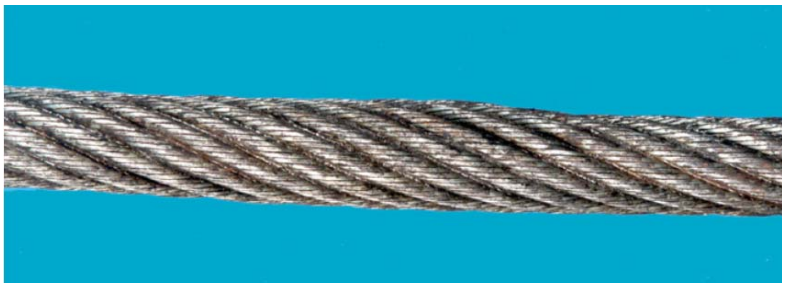
5: d=nominal diameter of rope.

Table 4-4: Typical examples of wire rope damages

No.	Description	Picture
1.	Crown wire breaks Usually caused by fatigue	
2.	Valley wire breaks	
3.	Localized grouping of broken wires	 
4.	Wire protrusion	

No.	Description	Picture
5.	Core protrusion-single-layer rope	
6.	Local reduction in rope diameter-sunken strand	
7.	Strand protrusion/distortion	
8.	Flattened portion	
9.	Kink(positive)	

No.	Description	Picture
10.	Kink(Negative)	
11.	Waviness	
12.	Basket deformation	
13.	External wear	
14.	External corrosion	

No.	Description	Picture
15.	Internal corrosion	
16.	Protrusion of inner rope of rotation-resistant rope	
17.	Local increase in rope diameter due to core distortion	
18.	Kink	

No.	Description	Picture
19.	Flattened Portion	

	In case of wire rope abnormalities, the maker of the wire rope should be determined. A colored tracer thread is embedded in the core of the rope. The manufacturer can be found by this tracer thread.
	We are using right hand lay ropes on both right and left handed drums  **Left hand lay**  **Right hand lay**

4.3.6 Procedure of wire rope inspection

4.3.6.1 Inspection procedure of main hoist rope

Table 4-5: Manpower and tools for the inspection of main hoist wire rope

Description		Quantity	Notes
Main power	Operator	1	With radio
	Examiner	1～2	With radio
	Supervisor	1	With radio
Tools	Duster cloth	Some	For clean
	vernier caliper	1	For measurement
	screwdriver	1	For checking internal of rope
	Chalk/tape	1	For marking
	Inspecting form	1	For recording
	Pen	1	
	Others		Other special tool for rope inspection.

The minimal manpower and tool for the inspection of main hoist rope are given in Table 4-5. The boom shall be leveled and the crane shall be switch to maintenance mode before inspection. Use the “Mode of rope changing” or using the emergency drive to run the main hoist drum.

Examination may be according to following steps:

1) Move the trolley to the outreach with the spreader at its highest position. When the trolley locates at the boom tip, lower the spreader at slow speed to as close to the water surface as possible. Mechanic(s) can stay at the landside of the main hoist drum in the machinery house to perform rope inspections. This step aims to check the wire ropes wrapped on the drum when the spreader is lowered. See position ① in Figure 4-20. Pay close attention to the moving drum while inspecting.
2) Move the trolley to the backreach at slow speed. Examiner can sit at the waterside corner platform of the trolley to check the main hoist wire ropes. This step aims to check the wire ropes between the outreach and the backreach. See position ③ in Figure 4-20. The Examiner must wear safety harness, and install safety chains near the rope inspection platform and pay special attention to any potential crush hazard. The spotter should stand on the operator’s cab to monitor the inspector’s clearance.
3) Lower the spreader to the ground
4) Hoist the spreader at slow speed, Mechanic(s) can stay on the landside corner platform of the

trolley to check the main hoist wire ropes. See position ④ in Figure 4-20.

5) Move the trolley forward to the park position while a mechanic standing on the landside trolley rope inspection platform. See position ⑤ in Figure 4-20.

6) The dead wraps of rope on the drum with the spreader at wharf level must be also inspected

Once the above-mentioned procedure is completed, the full length of main hoist wire rope has been inspected

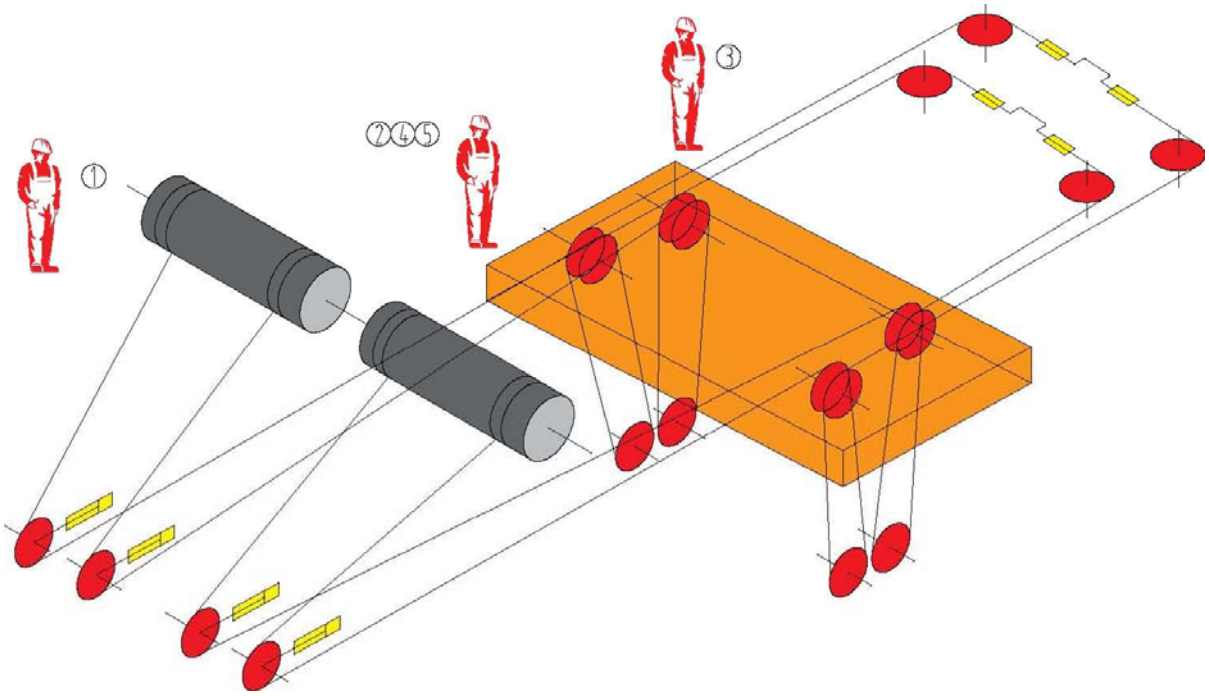


Figure 4-20：Inspection of main hoist rope

	**Danger of falling!**
	**Danger of tripping!**
	**Keep hands clear of moving parts!**
	**The crane must be in maintenance mode for this task. When a person stands on the trolley, he must wear safety harness, and install safety chains near the rope inspection platform. If hands have to be on rope for inspection, they must always be away from the pinch point. An additional person must act as a spotter between the operator and the mechanic(s) inspecting the rope.**

Headblock Sheaves

4.3.6.2 Inspection of trolley towing wire rope

	**The crane must be in maintenance mode for this task. When a person stands on the trolley, he must wear safety harness, and install safety chains near the rope inspection platform. If hands have to be on rope for inspection, they must always be away from the pinch point. An additional person must act as a spotter between the operator and the mechanic(s) inspecting the rope.**
	**Danger of falling!**
	**Danger of tripping!**
	**Keep hands clear of moving parts!**

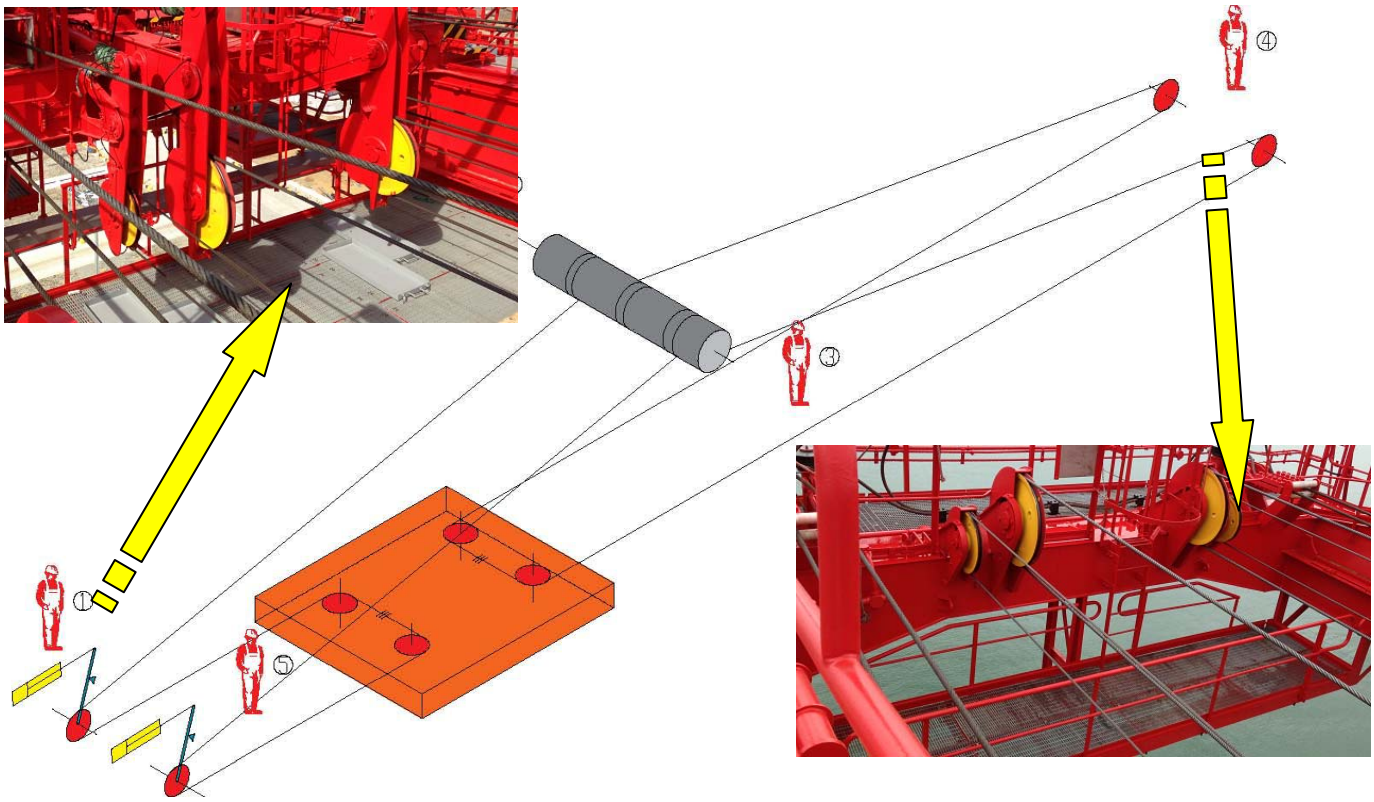


Figure 4-21：Inspection of trolley towing rope

Prepare the manpower and tool according to Table 4-5. The boom shall be leveled and the crane shall be switch to maintenance mode before inspection. Use the “Mode of rope changing” or using the emergency drive to runn the trolley drum.

Following steps should be followed for the inspection.

1) Move the trolley to the back reach at slow

speed.

2) Move the trolley from rear of backreach to the boom tip at slow speed. Examiner(s) can stay at the platform (as indicated in picture above) near the backreach trolley deflector sheaves for rope inspection. See position ① in Figure 4-21. Spotter should stand at backreach with a radio ready to push the E-Stop.

3) Move the trolley from boom tip to back reach at slow speed. Examiner(s) can stay at the waterside of the trolley drum in the machinery house for ropes inspection. See position ② in Figure 4-21. Spotter should stand at emergency drive with a radio ready to push E-Stop.

4) Move the trolley to the backreach at slow speed. Examiner(s) can stay at the boom tip platform for rope inspection coming off the lower sheave. See position ④ in Figure 4-21. Spotter should stand at boom tip E-Stop with a radio.

5) With the trolley at the full backreach position and without trolley motion, Examiner(s) can stand on top of the cab to inspect the ropes on the landside of the trolley. See position ⑤ in Figure 4-21

6) Examine the rope remaining on the drum.

7) Once the above-mentioned procedure is completed, the full length of trolley towing wire rope has been inspected

4.3.6.3 Inspection of boom hoist wire rope

	**Danger of falling!**
	**Danger of tripping!**

	**Keep hands clear of moving parts!**

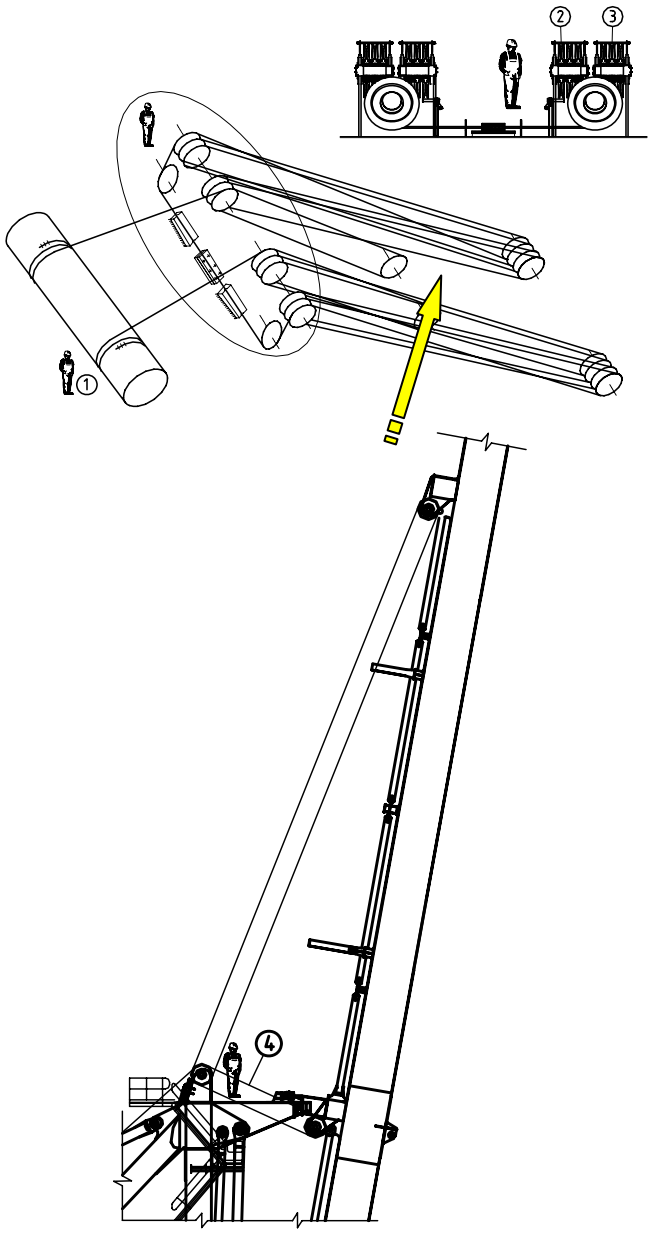


Figure 4-22：Inspection of boom hoist wire rope



Figure 4-23：Boom hoist sheave units at the top of A frame

The minimal manpower and tool for the inspection of main hoist rope are given in Table 4-5. The boom shall be leveled and the crane shall be switch to maintenance mode before inspection. Use the "Mode of rope changing" or using the emergency drive to run the drum.

Following steps shall be followed for the inspection

1) Hoist the boom at slow speed to its maintenance position. Mechanic can stay on the landside trolley girder support beam for rope inspection and cover as much of the wire rope as possible, and mark the end of inspected segment. See position ① in Figure 4-22.

2) Lower boom at slow speed to its operation position. Mechanic can stay on the top cross tie beam near 1st sheave (count from the center line of the boom). The wire rope reeving exiting the 1st sheave can be inspected. See position ② in Figure 4-22.

3) Hoist the boom and then lower it at slow speed to its maintenance position. Mechanic(s) can stay on the top cross tie beam to inspect the rope reeving through the 5th sheave (count from the center line of the boom) while the boom is lowering. See position ③ in Figure 4-22.

4) Inspect the remainder of the wire rope from the top cross tie. See position ④ in Figure 4-22.

Once the above-mentioned procedure is completed, the full length of boom hoist wire rope has been inspected.

4.4 Wire rope lubrication

	**It has been proven that a well lubricated wire rope will last four times as many working hours as a non-lubricated wire rope.**
	**Always use wire rope manufacturer's recommended lubricant!**

4.4.1 Importance of lubrication

The importance of periodic lubrication will never be overstressed. Mainly they are:

1) A wire rope is a machine, having many moving parts. Each time a rope bends or straightens the wires in the strands of the rope must slide on each other. Thus, a film of lubrication should be present on each moving part. When exposed to the same test condition a well-greased rope showed four times as many working cycles as a non-greased rope.

2) A second important reason for lubricating the wire rope is that it keeps away moisture which can cause interior corrosion under simultaneous bacterial proliferation in the fiber core. The lubricant also prevents deterioration of the fiber core. A rusty rope is a liability, since there is no means of visual inspection, which can determine the remaining strength of a corroded rope.

4.4.2 Selection of lubricant

the main lubricants for wire ropes used in a contaniner terminal are below:

1) Grease: Mainly is lithium soap base grease.

Sometimes are mixed with mineral oil for better penetrability.

2) Special lubricant for rope such as : ESSO SK(semi-liquid), FUCHS Ropelife (Grease),

3) Special synthetic base oil for rope such as: Ultrachem Chemlube 812(synthetic ester base oil), synthetic ester base oil supplied by PAO

The grease that is one kind of protection of coving to rope may be most wildly used for the lubrication of the rope traditionarily because of convenience for using, economical efficiency and efficiency also good effectiveness. Compared to the lubrication oil that is liquid and has better penetrability, the grease are more easily to be tainted with foreign grains in larger size such as dust, slag or grits. Those hard grains may also be pushed inside the wire rope and cause direct damages to rope. On the other hand, Oil lubrication also has better performance in resist the moisture, salt spray for the interior of rope than grease lubrication. Additionally grease lubrication is also making the visual inspection more difficult. Please reference section 2.6.2 for the lubricant used in this crane before it left the factory.

Below are some main requirements to the lubricant of the wire rope used in STS crane

1) Better antioxidant capacity.

2) The better capacity to insulate the wires from moisture and salinity. That means the lubrication shall have a strong capacity of adsorbing and binding with the wires.

Usually ester oil has a better performance in Iron-binding capacity, permeability, film strength and water resistant. The user may determine the lubrication according to the actual condition of different terminal.

4.4.3 The method of lubricant

Lubricant may be applied with an oil can, brush or any effective method. The objective is to apply a uniform coating to the entire length of the wire rope. The detail scheme shall be make to assure the full length of wire rope can be lubricated. The wire rope may be lubricated by manually or Semiautomatically or automatically. The typical manual lubricating is shown in Figure 4-24.

A semiautomatic method is shown in Figure 4-245. This method may be more convenient and effective. The lubricant is easily to go inside the wire rope by the pressed air. Frequent and not too excessive lubrication of the wire ropes produces better results than only occasional but rich lubrication. As soon as the grease film vanishes in some areas of the wire rope, it has to be re-greased.

Greasing has to be done only when the wire rope is completely dry. Otherwise, the existing moisture will be contained in the rope and the corrosion be increased from the inside. Used wire rope should be cleaned with a wire brush to remove any accumulation of dirt, grit, rust, or other foreign matter. The purpose is to remove all old lubricant and foreign matter from the valleys between the strands and from the spaces between the outer wires. This gives newly applied lubricant ready penetrate into the rope.

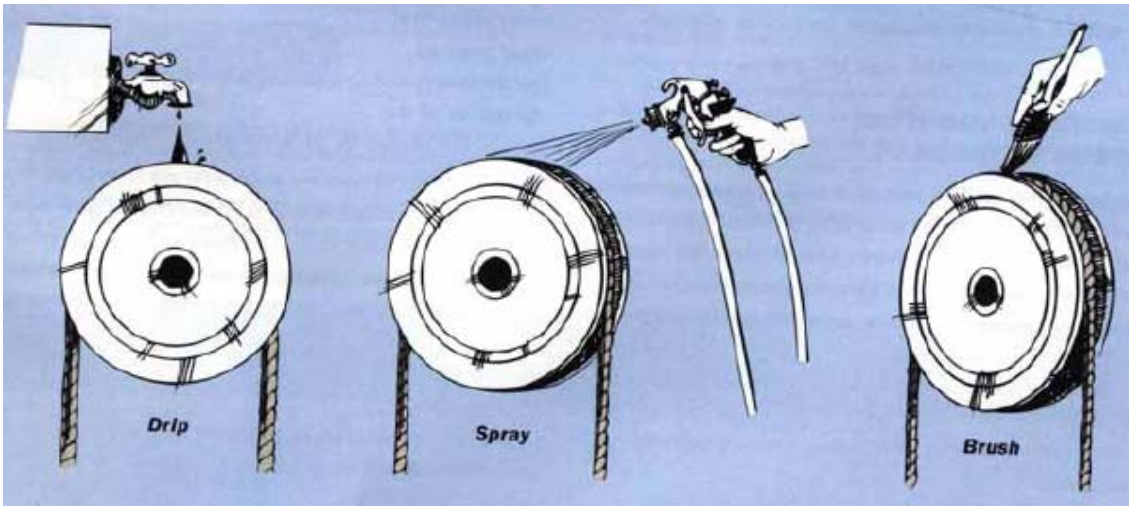


Figure 4-24：Method of manual lubricating

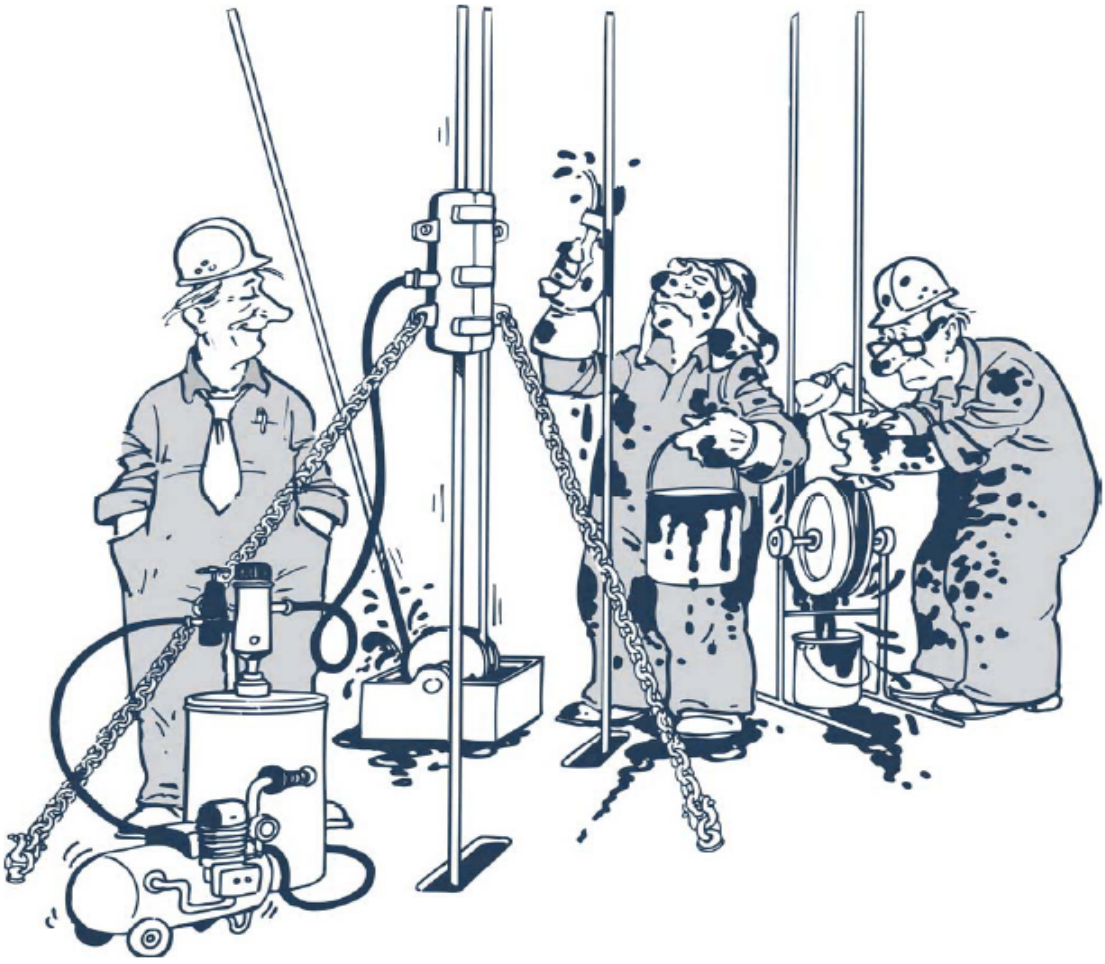


Figure 4-25：Semi-auto Pneumatic lubricating

4.4.4 Lubricating of main hoist and trolley towing wire ropes

Introduction of wire lubrication below are based on the method of semiauto pneumatic lubricating. The procedure is similar if the other methods are used.

During the whole procedure, the boom shall be level and the crane must be in maintenance mode.

The necessary manpower and tools are shown in Table 4-6.

Table 4-6: Manpower and tools for the inspection of main hoist wire rope

Description		Quantity	Notes
Main power	Operator	1	With radio
	Examiner	1～2	With radio
	Supervisor	1	With radio
tools	Duster cloth	Some	For clean
	vernier caliper	1	For measurement
	screwdriver	1	May be use to check internal of rope
	Chalk/tape	1	For marking
	Inspecting form	1	For recording
	Pen	1	
	Others		Other special tool for rope inspection.

1) Ensure that the boom is lowered and the crane is in Maintenance Mode.

2) Move the trolley to the suitable position such as boom tip so the

3) Select a proper position to start the lubricating such as drum. Parts of the rope can be lubricated by rotating the drum slowly. Trolley is also a necessary good position to lubricate. Most of the remaining rope can be lubricated by moving the trolley slowly. There are always some dead ropes that could not reach on trolley or can not go into the drum. You have to manually lubricate it by the bushes with extension rods. Anyway, the full length of the rope shall be lubricated.

	**Danger of falling!**
	**Danger of tripping!**
	**Keep hands clear of moving parts!**

4.4.5 Lubricating of boom hoist wire ropes

There are some “dead ropes” between the sheaves on top of A frame and sheaves on boom tip and could not reached by normal access. These ropes are long and very difficult to be lubricated by the normal methods. To lubricate those ropes, additional tools listed in Table 4-7 are necessary besides the manpower and tool listed in Table 4-6.

Table 4-7: Additional Tools to lubricate the boom hoist rope

Description		Quantity	Note
tools	Fiber rope	140m	
	Air hose in reels	65m	

The recommended procedure to lubricated the boom hoist ropes are below:

1) Boom shall be level and crane shall be in maintenance mode.

2) Operate boom in the boom operation cabin

3) Clamp the lubricator onto the rope above the landside trolley girder support beam near the machinery house. See ① in Figure 4-26. Turn on the lubricator and hoist the boom to maintenance position at slow speed.

4) Clamp the lubricator on the lead-line rope just landside of the sheave group on the top cross

tie beam of A-frame which leads to drum, and connect the lubricator air hose to the air quick disconnect. See ② in Figure 4-26.

5) Turn on the lubricator and lower the boom at slow speed until the freshly lubricated rope is at the waterside of the lubricator

6) Clamp the lubricator on the rope reeving from 4th sheave (count from the centerline of boom) on the top cross tie beam to the third sheave (count from centerline of the boom) on the boom sheave group. See ③in Figure 4-26

7) Turn on the lubricator and lower the boom to the operating position at slow speed.

8) The last segment of wire ropes can be lubricated as shown in Figure 4-27. Connect the pneumatic lubricator with long hose in reels. Installed the sleeves of lubricator on the ropes. Then secure all the parts of lubricator connected with the sleeves by chain in case of falling down during the moving of lubricating later.

9) The lubricator is towed with a fiber rope at both ends. The fiber rope goes around a temporary sheave as shown in Figure 4-27. One mechanic can pull the rope on the walkway of boom while another mechanic hold other fiber rope at top of A frame, so the lubricator can move along the last segment of the wire ropes slow and then the rope to be lubricated simultaneous.

10) The dead wraps on both sides of the drum should be lubricated using a brush

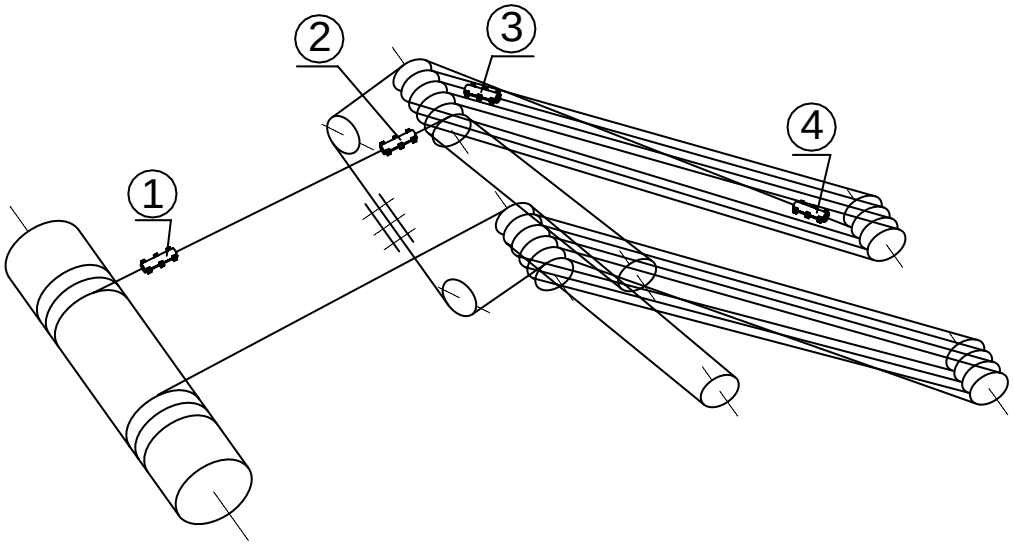


Figure 4-26：Operating position for boom hoist rope lubricating

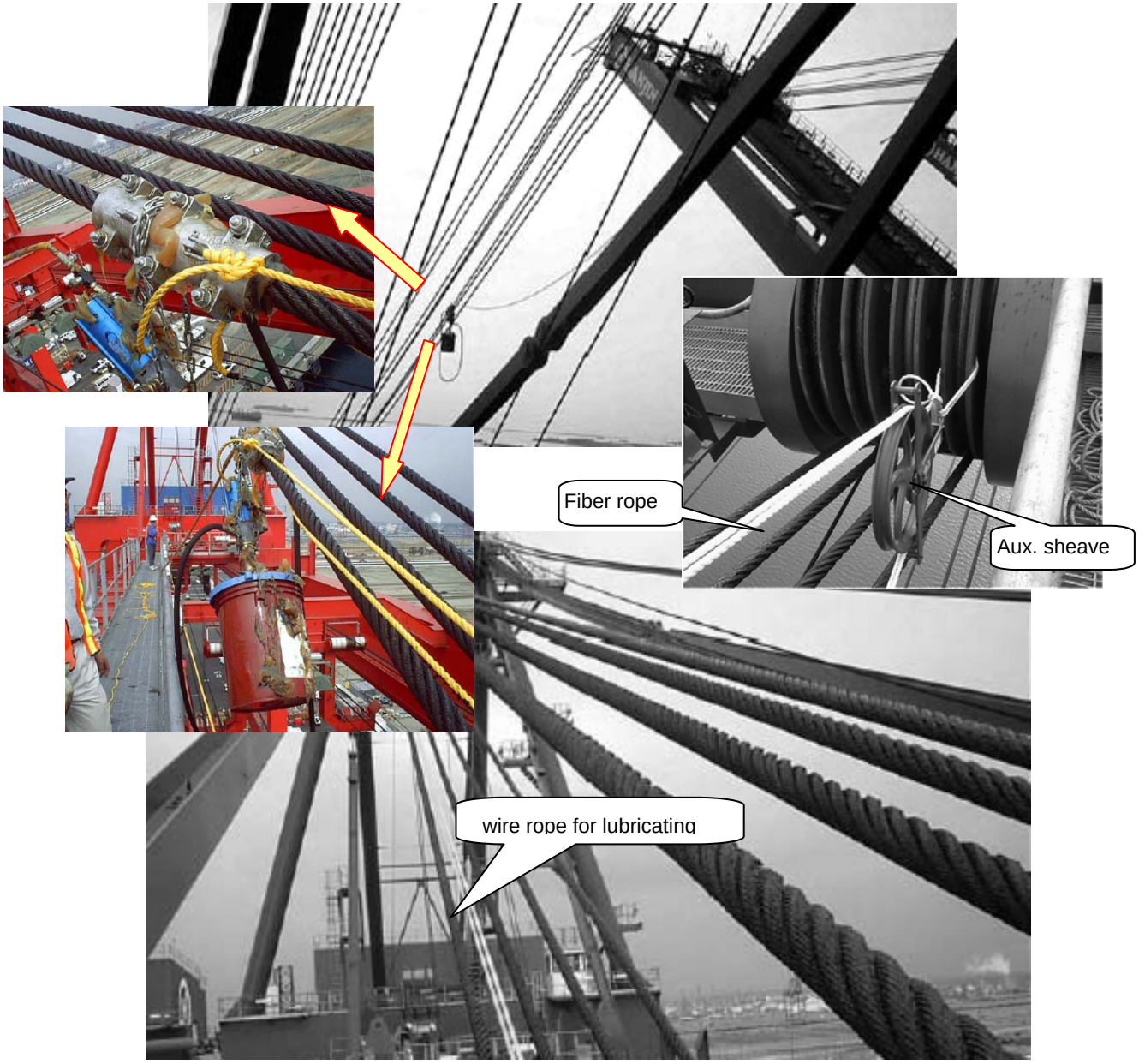


Figure 4-27: The lubricating of “dead rope” of boom hoist

	**Danger of falling!**
	**Danger of tripping!**
	**Keep hands clear of moving parts!**

4.5 Rope re-reeving

The ropes reach the discard criteria shall be replaced by new one. The rope can be changed by the rope re-reeving device inside the machinery house. Please see section for the instruction of this device. The manpower and tools needed for the rope re-reeving are given in Table 4-78. The wire rope type and size are given in Table 4-9. Reference draiwngs are also given in

Table 4-10.

Table 4-8: Manpower and tools needed

Description		Quantity	Notes
Manpower	Mechanics	5	With radios. At least one person is qualified as a operator.
	Supervisor	1	With radios
	Duster cloth	Some	For cleaning
tools	Chisel	1	
	Torque wrench	1	
	Wrench	One for each size	M16,M20,M24, M30, M36
	Support for reel	1	Provided for maintenance staff
	Rope clamp	5	Special made and work with chain block
	2～3t chain block	5	Two chain block (3t) for boom hoist rope changing
	Wood pad	1	Round and 80mm in diameter
	Rubbers, wood plate, snap block	Some	
	Wires, steelplate	several	Secure the wire, plate, slap block, or tightening rope end.
	Grinder or rope cutter	1	Cut wire rope
	pincers	1	To colligate rope end
	Chinese fingers	1	Connect the old and new ropes
	electric welding machine	1	Welding temporize parts
	Chalk/ punched tape	1	For marking
	Note book and pen	1	For recording
	Drawings and instructions of re-reeving	1set	

Table 4-9:：Rope type and length

Description	Note
Main hoist rope	Type: TOKYO IWRC 6xWS(36)-B-1770N/mm2, right lay Dia: 31.5mm Min Break load：696KN Length: 672m, totally 2/ crane
Trolley towing rope	Type: TOKYO IWRC 6xP.WS(36)-B-1770N/mm2 , right lay Dia：25mm Min Break load：484KN Length：（523+351）m, each 1/crane
Boom hoist wire ropes	Type: TOKYO IWRC 6xP.WS(36)-B-1770N/mm2 , right lay Dia.：35.5mm Min Break load：975KN Length：1480m, totally 1/crane

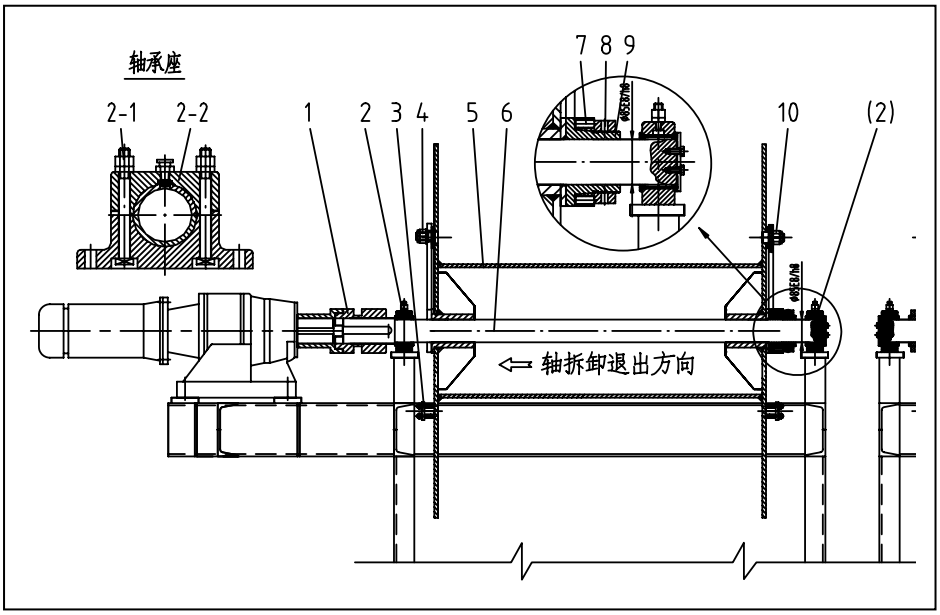
CatenaryTrolley towing rope	Type: TOKYO IWRC 6xP.WS(36)-B-1770N/mm2 , right lay Dia：16mm Min Break load：173KN Length：（182+210）m, each 2/crane

Table 4-10: Reference drawings

Drawing No.	Description
J526A15	Rope re-reeving system
WRR04A-J526A	Rope re-reeving device
J526A2000	Main hoist rope reeving system.
J526A2100	Trolley towing rope reeving system.
J526A2200	Catenary Trolley towing rope reeving system.
J526A2300	Boom hoist rope reeving system.

	**Barricade the area under the crane where falling parts or rope may harm person and vehicle. Don't allow the person other than concerned to enter the procedure area.**
	**Keep away from suspended loads!**
	**Danger of falling!**
	**Danger of tripping**
	**Keep hands clear of moving parts!**
	**Pay attention to the rotating direction of reels! The rotating direction(top of reel) shall towards the handle of the bind brake.(match the direction of brake torque)**

4.5.1 Disassembling or assembling of rope reel



1. Clutch 2. Bearing seat 3. Rope clamp 4. Reel securing blot 5. Driving reel 6. Driving reel shaft 7. Inner lock sleeve 8. Outer lock sleeve 9. Round nut 10. Securing bolt of inner lock sleeve 2-1 Securing bolts of upper cover of Bearing seat 2-2 Upper cover of Bearing seat

Figure 4-28: Rope Re-reeving device

See Figure 4-28 for detail construction of the rope re-reeving device. The procedures of installation or disassembling of the reel are as below:

1) Disconnect the clutch by rotating the handle. Loose and take out the securing bolts of upper cover of bearing seat(2-1, 2-22 in Figure 4-28)

2) Move the support on the suitable position inside the machinery house. Then Lift the reel(driving side) together with the shaft by service crane and put it on the support.

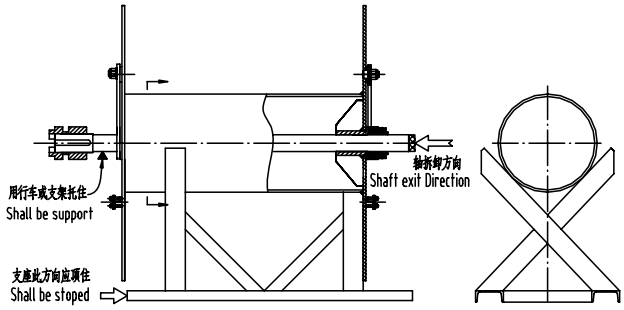


Figure 4-29: Step 2、6

3) Loose and take out the round nuts of item 9.



Figure 4-30: Step 3

4) Take off the securing bolt of inner lock sleeve.



Figure 4-31: Step 4

5) Knock slightly around the outer/inner lock sleeve by chisel.



(a)



(b)

Figure 4-32：Step 5

6) Put the wood pad on the end of shaft and then hit it by hammer. When the shaft is hit into inner of the reel, a short shaft to be put at the end of shaft is necessary so the shaft can be hit out continuously until the whole shaft are taken out. Another end o shaft shall be supported by the support. Also refer to Figure 4-28:

7) Move the shaft to a proper position such as the bearing seat of the reel. Lift the reel support out of the machinery house

8) Installation of the reel is just in reverse.

4.5.2 Procedure of main hoist rope re-reeving

The following procedure is assumed that the rope which needs to be changed is the outer one. If the rope need to changed are another one, the procedure are just similar.

1) Move the boom to horizontal. Hang a warning signs to warning ships if necessary. Parking the trolley and then hoist down until the spreader

on the ground. Switch the switch on the control station inside the machinery house(Fig.4-33) to the position of "Enable".



Figure 4-33：Operating controls for rope re-reeving

2) Lift the reel with new rope into the machinery house and reel the wire rope on the idle drum.

3) Engage the clutch (item 1 in Figure 4-28) by rotating the handle.

4) Trim, list, skew to Zero position.

5) Secure the main hoist rope in front of the position of trim, list, skew device at rear of backreach (Position 1、2、1'、2' as shown in Figure 4-38. Also see Figure 4-34.



(a): Chain block



(b): Special rope clamp



(c): Securing point at backreach

Figure 4-34：Wire rope securing point and tools

6) Hoist down until only 3 wraps on the drum (about 12m). Then most ropes are hanging between machinery house and main hoist sheaves at rear of backreach. See Figure 4-38.

7) Loosing the securing bolts of rope clamps on drum and take off the rope clamps.

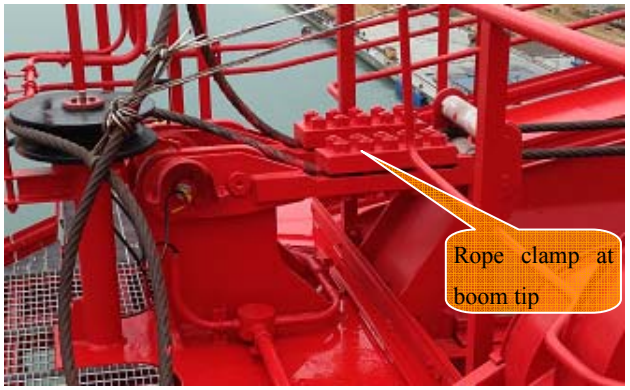


Figure 4-35：Step 7

8) Secure the wire rope which needs to be changed at driving reel side in front of the main hoist drum (Point 3 shown in Figure 4-38).

9) Loose and take off the wire rope clamp from the drum. Then take off the rope from the drum and pull it through the guider sheaves under service crane and reach to the driving reel and be secured on it. See Figure 4-39.

10) Loose the rope security at point 1, 3 as shown in Figure 4-38.

11) Operating on the rope re-reeving operating station as shown in Figure 4-33. Run the driving reel and take up the old rope slowly into the driving reel until the rope belly at backreach disappeared. Loosing the rope security at point 1', 3 as shown in Figure 4-38. Watch along the rope path especially the belly at backreach and rope passing the headblock and make sure rope does not jump out of the

sheaves or touch the steel structures directly.

12) Secure another rope end on left side drum on the position in front of main hoist drum (Point 3' as shown in Figure 4-39) by rope clamp and chain block.

13) Loosing the left side drum clips and pull the rope to go through the guider sheave to the idle reel. Connect the old rope and new rope with Chinese fingers or by welding.



Figure 4-36：Cable grip

14) Loose the point 3 as shown in Figure 4-39.

15) Run the driving reel again and pull the old rope slowly until 12m length new rope appears and go inside the machinery house. Watch along the rope path especially the belly and headblock make sure rope does not jump out of the sheaves or touch the structure.

16) Secure the right side rope in front of the drum (point 3 as shown in Figure 4-40).

17) Disconnect the old rope and new rope. Then pull the rope end to the main hoisting drum and install three Wraps (same as another old rope not to be changed). Install the rope clips on drum and tight it preliminary. Loose the point 3 as shown in Figure 4-40. Also see Figure 4-41.

18) Hoist up slowly. Main hoist drum will pull the new rope in. See Figure 4-41

19) Secure the point 1 as shown in **Figure 4-41**. Hoist down until 3 Wraps on the drum. Then there are totally three bellies at backreach. One of them is new rope belly.

20) Install the rope clamp at boom tip. Adjust the rope until the new rope has same catenaries as the other old one. Tight the bolt to 75% of the final torque

21) Secure the point 1' as shown in Figure 4-41 (driven reel side new rope). Then pull the rest of new rope from the reels manually and then there are four bellies at backreach. Make sure all four bellies along the full length of rope are the same or can be adjusted by chain block.

22) Secure the point 3' as shown in Figure 4-41.

23) Take out the rope end from the driven reel and move it to the drum and install 3 Wraps (Same as another old rope has not been changed) on it and fix it by rope clips. See fig 4-9. Then secure it on the drum by rope clamp. See Figure 4-42.

24) Loosing the point 3' as shown in Figure 4-41. Hoist up slowly. Loosing the point 1、2、1'、2' as shown in Figure 4-41 until the spreader start to leave ground. Please reference Figure 4-42.

25) Check if the position of spreader is correct and when it hoist up to max check if there are about 1~2 empty rope grooves on left on the drum. Or the ropes need to be adjusted. Sometimes the trim, list & skew device may also need to be reset.

26) Cut the remaining rope and tight all the bolts according to Table 4-11.

27) Runing the main hoist up and down slowly and check if the control system is working properly such as: the position of Cam. Correct every abnormal phenomenon immediately.

28) Lubricate the new rope. This can also be done during the rope changing simultaneous.

Table 4-11: Bolt tighten torque

Location/size	Toque

Boom tip rope clamp/M27	780N m
Rope clips on drum/M30	900 N.m

Watch along the rope path especially the belly at backreach and rope passing the headblock and make sure that rope does not jump out of the sheaves or touch the steel structures directly.

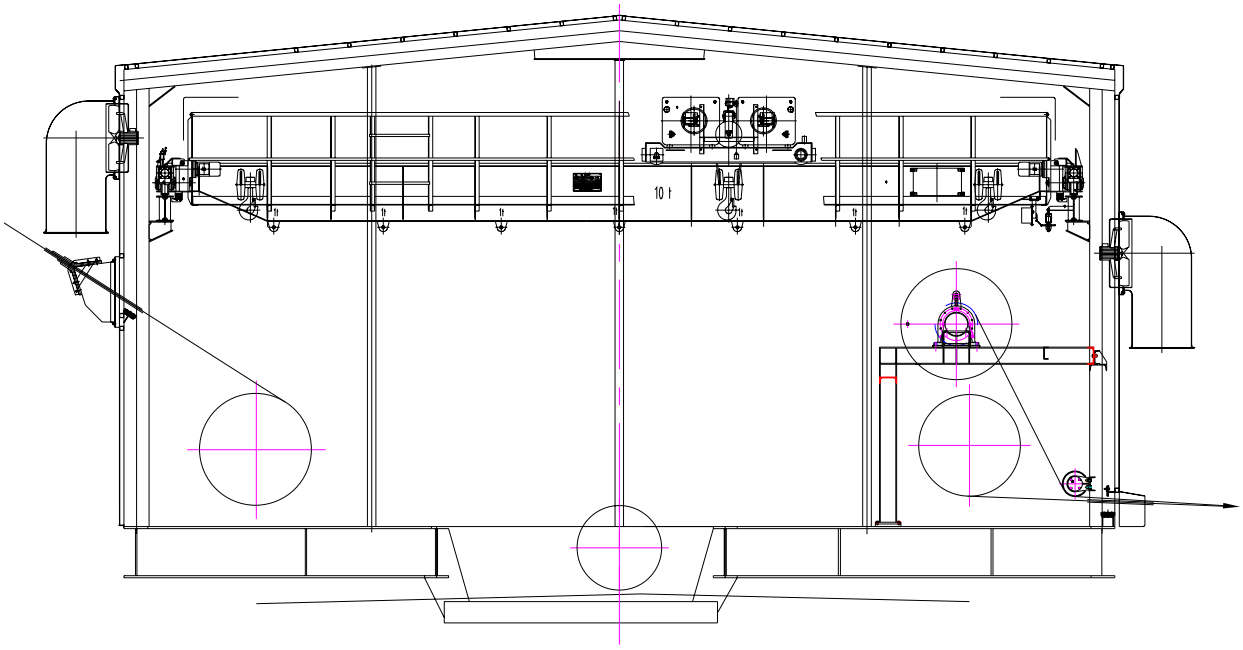


Figure 4-37：Rope path inside machinery house

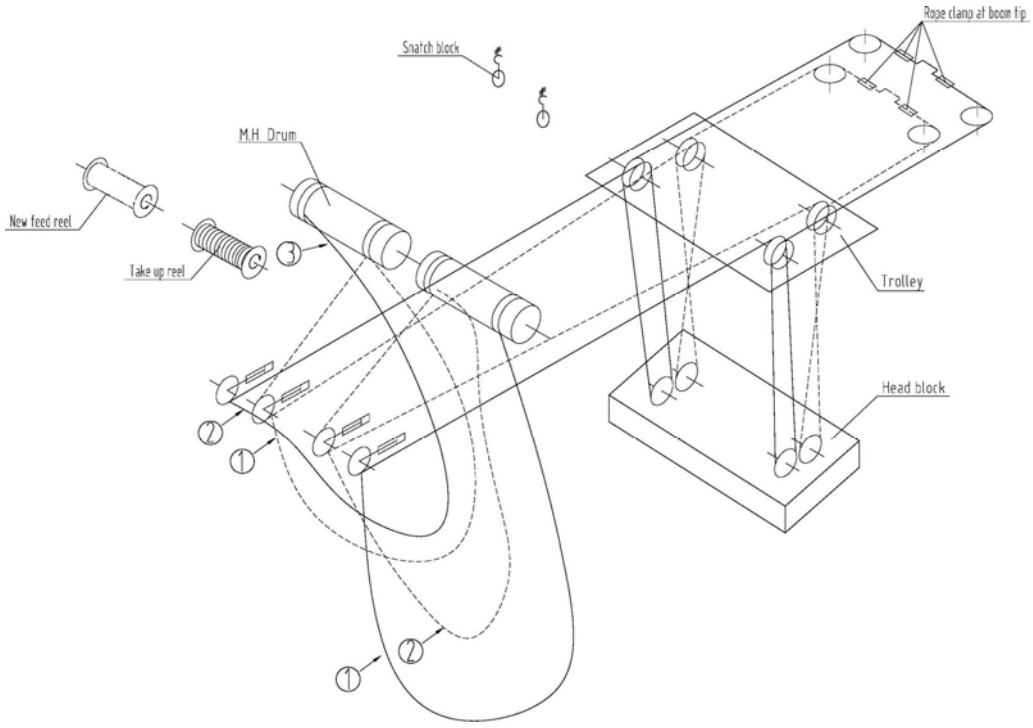


Figure 4-38：Step 6、7

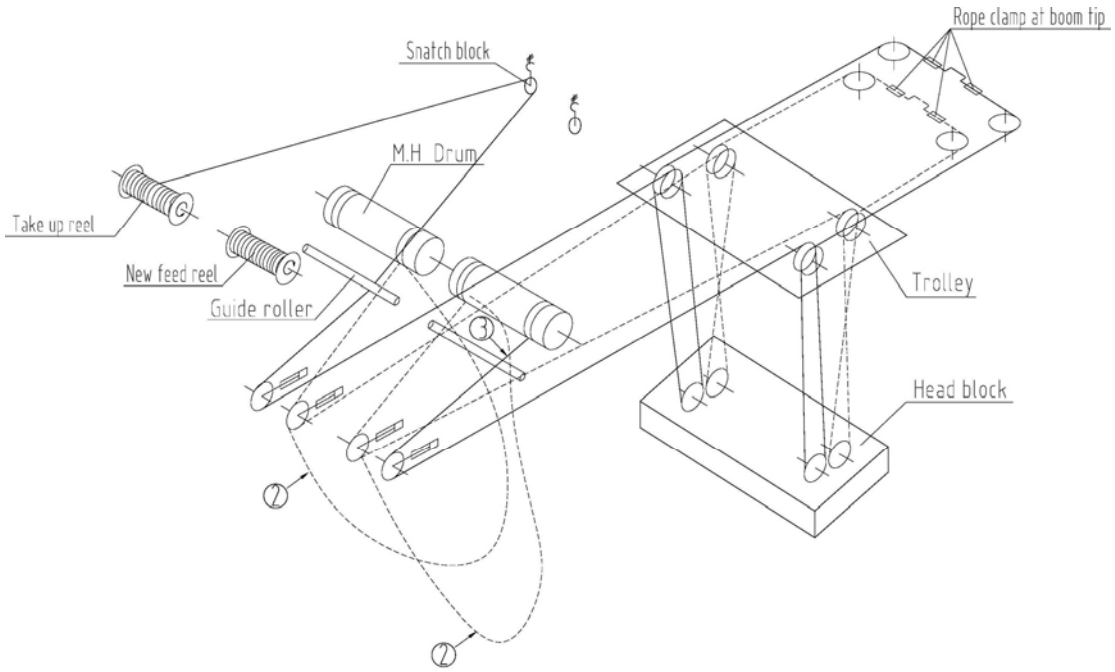


Figure 4-39：Step 9、10、11、12

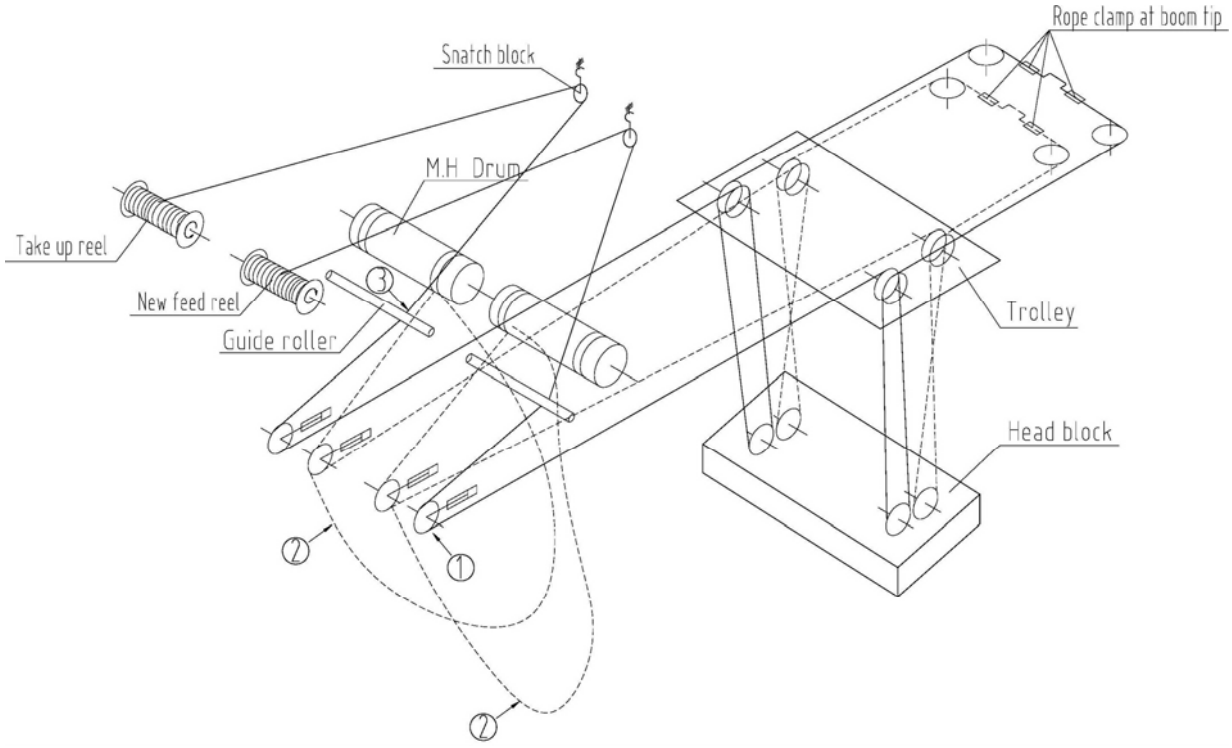


Figure 4-40：Step 13、14、15、16

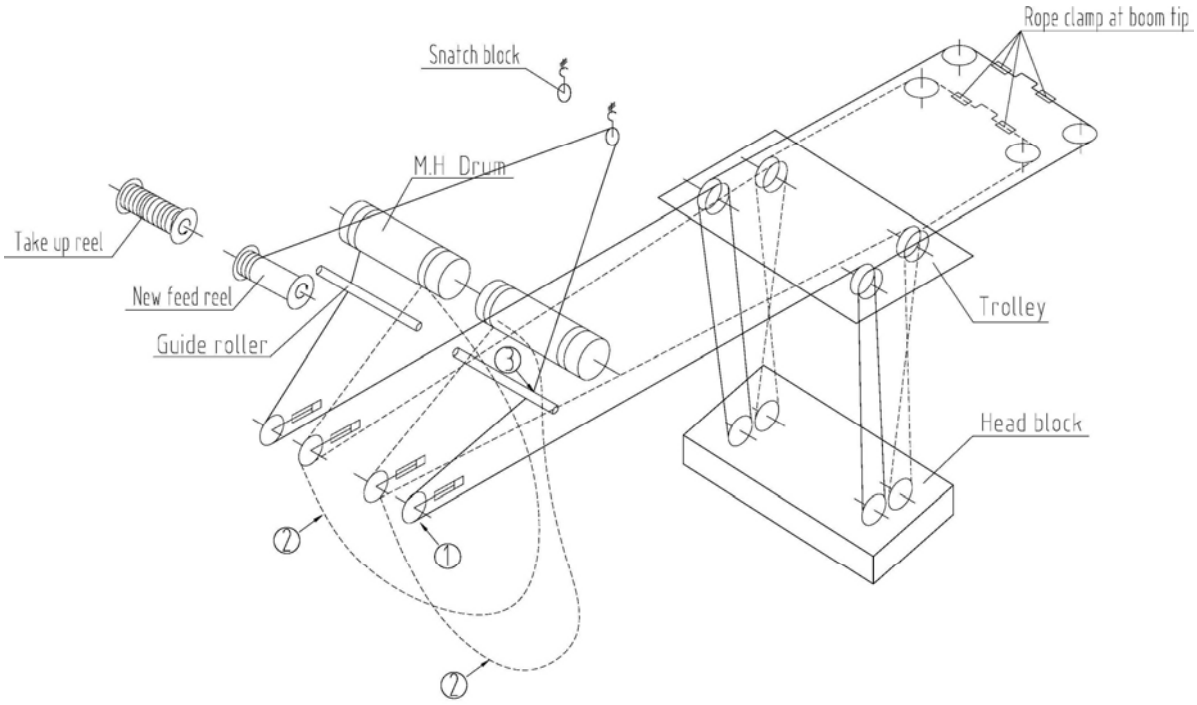


Figure 4-41：Step 17、18、19、21、22

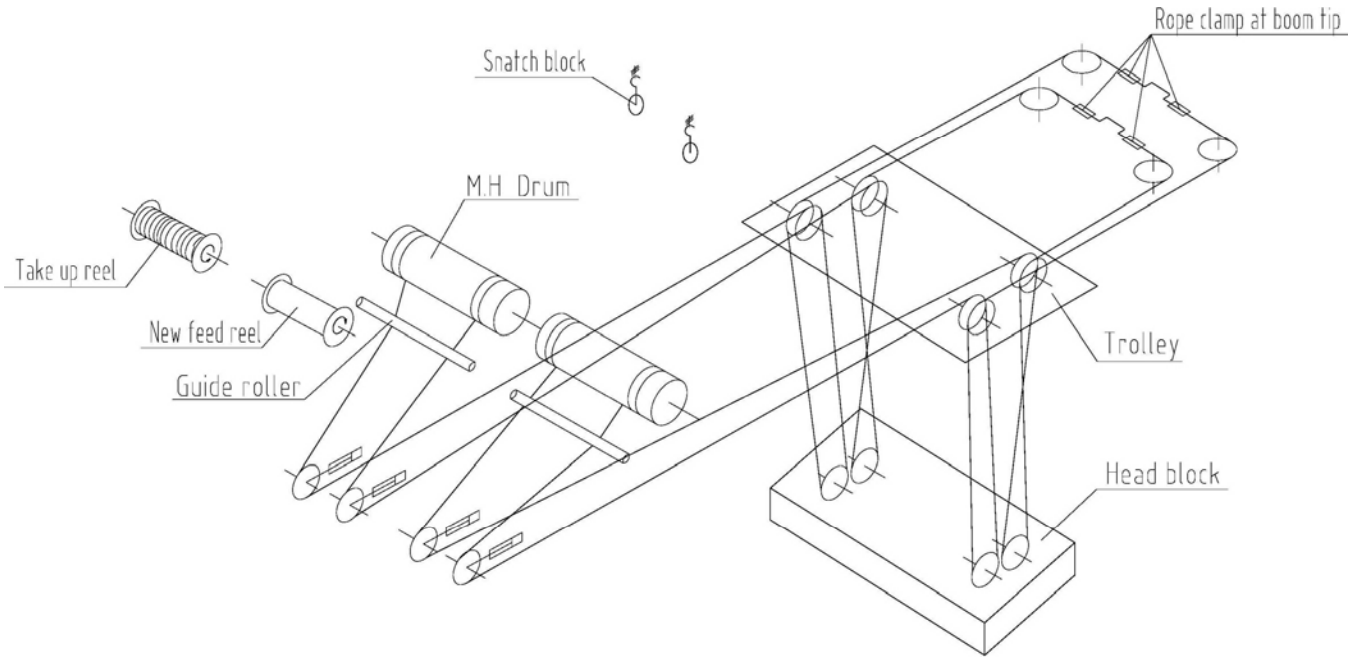


Figure 4-42：Step 23、24

4.5.3 Re-reeving of Trolley towing rope

Detail procedurea of re-reeving of trolley towing rope are below. It is assumed that the rope which needs to be changed is at seaside. The procedures are similar if the other rope needs to be changed.

1) Move the trolley to full backreach. Switch the crane to maintenance mode. See Figure 4-33.

2) Loose the rope clamps on trolley as show in. Figure **4-43**.



Figure 4-43: Step 2

3) Release the hydraulic pressure of trolley rope tension by opening break valve. See Figure 4-44.

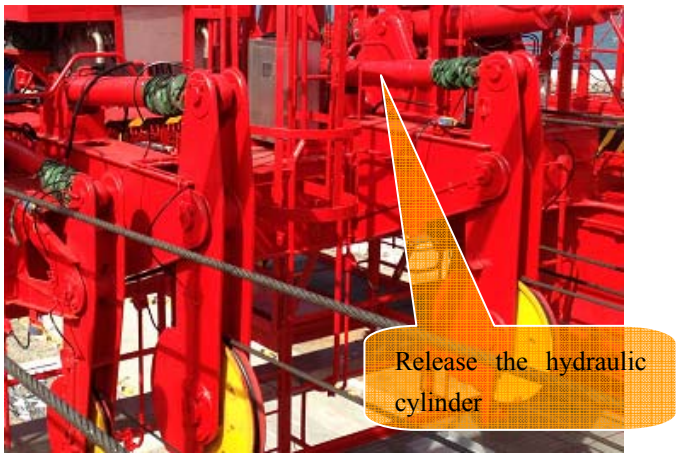


Figure 4-44: Step 3

4) Install two snatch block under service and two guider sheaves (Usually be put on the corner room inside the machinery house) in front of the trolley drum as shown in Figure 4-46.

5) Move the reel with new ropes into the machinery house and reel the ropes on the idle side (pay out side). At the same time. The empty reel is on driving side.

6) Clamp the rope in front of the drum(point 1，1' shown in Figure 4-47). Also See Figure 4-45.

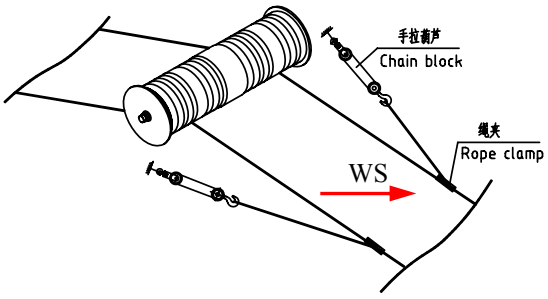


Figure 4-45: Step 6

7) Loose the seaside rope end securing clips on drum. Then pull the rope end (the rope need to be changed) to go through the guide sheaves and to the driving reel. Run the reel and tight the rope a little bit. See Figure 4-47.

8) Connected the old rope and the new rope, run the reel and tight the rope a little bit, See figure 4-48.

9) Release point 1 as in figure 4-48.

10) Drive the reel again until about 10m of new rope go into the machinery house, control the reel speed use the brakes. Watch and make sure that the ropes do not jump off the sheaves and that there is no damage to the rope along its running path, refer figure 4-49.

11) Secure the point 1 shown in Figure 4-50. Disconnect the old rope and new rope. Then install both rope ends at waterside on the drum by rope clips .Make sure the wraps on drum are same.

12) Adjust catenaries of the rope and make the two sides of the new ropes on the seaside are the same. Tight the rope clamp on trolley.

13) Secure point 2 as in figure 4-50, remove the rope ends of the new rope from the idle reel, transferring it to the drum. Make sure the loops are 7(trolley in the end of the backreach). Then release the rope clamp there.

14) Reset the pressure of trolley rope tension by

opening the break valve. Check if the ropes of two sides are equally. The arm of tensioner shall also be in vertical position. Or the rope need to be adjusted such as adjusting rope length at both sides of clamps on trolley or adjusting rope length on drum after releasing the pressure of trolley rope tension.

15) Run the trolley slowly to check if everything (including all the limit switches) is Ok.

16) Lubricate the new rope. This can also be done during the rope changing simultaneous.

Table 4-12: Bolt tighten torque

Location/size	Toque
Rope clamp on trolley/M24, 8.8s	420Nm
Bolts securing rope clips on drum/M24, 8.8s	420 Nm

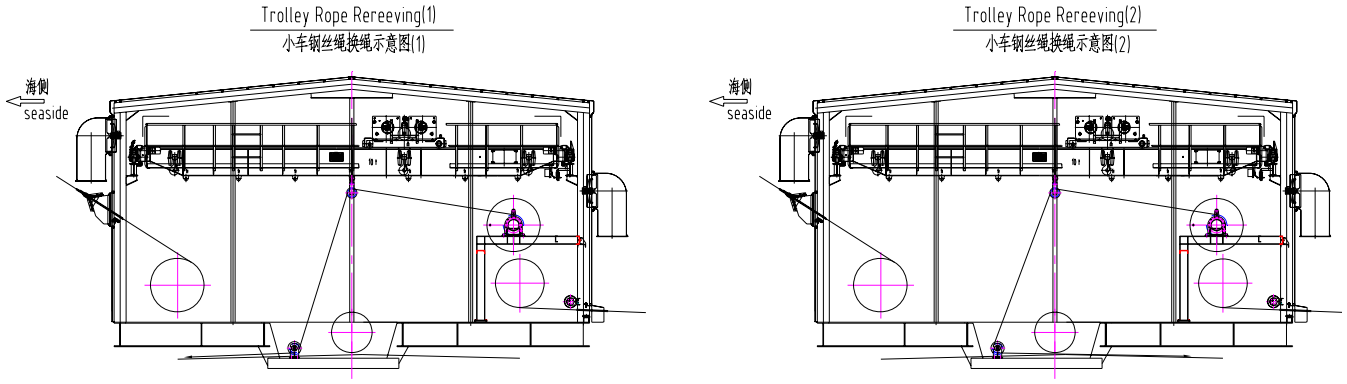


Figure 4-46: Arrangement of guider sheaves inside machinery house

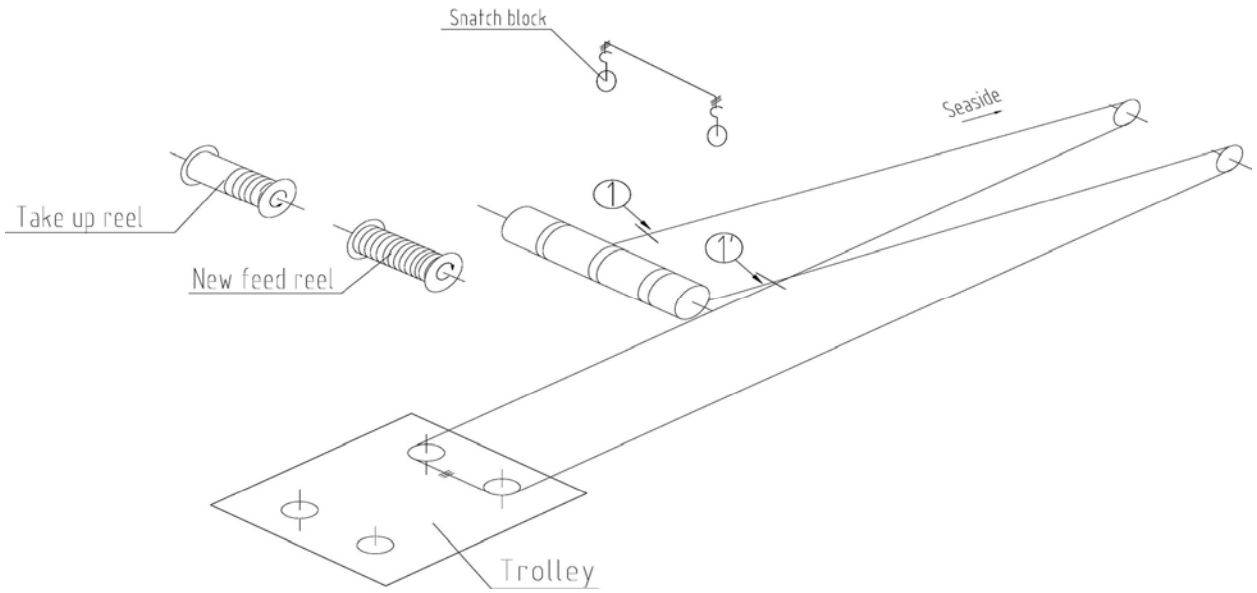


Figure 4-47: Step 6,

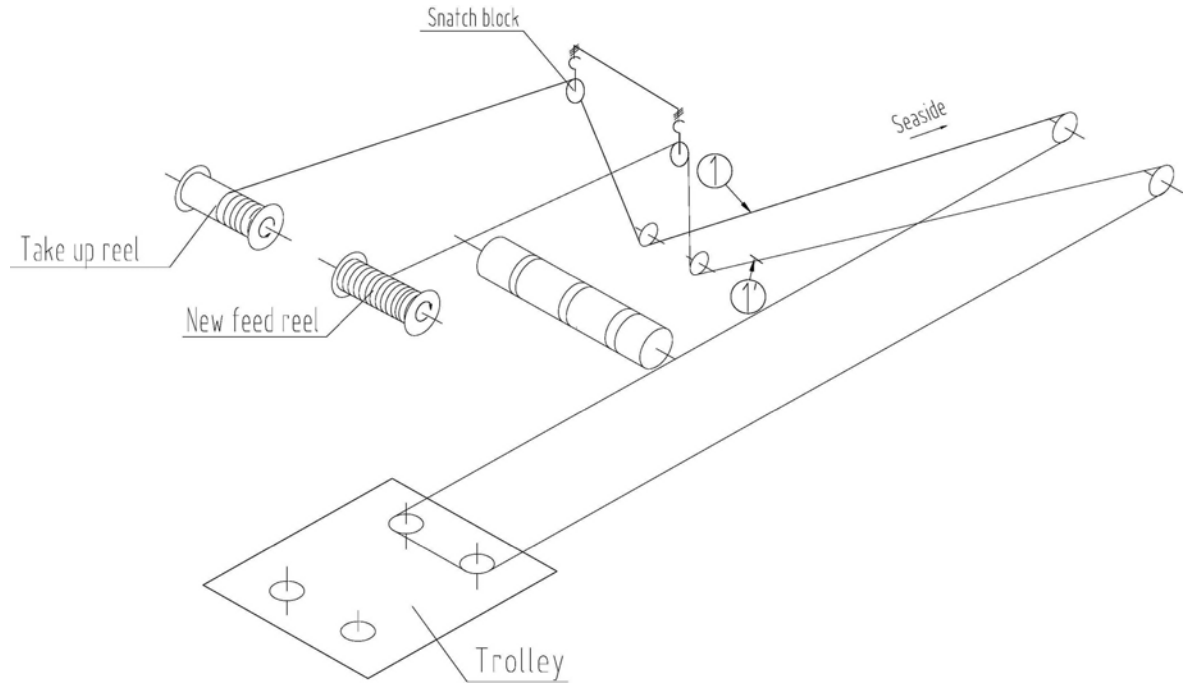


Figure 4-48: Step 7,8,9

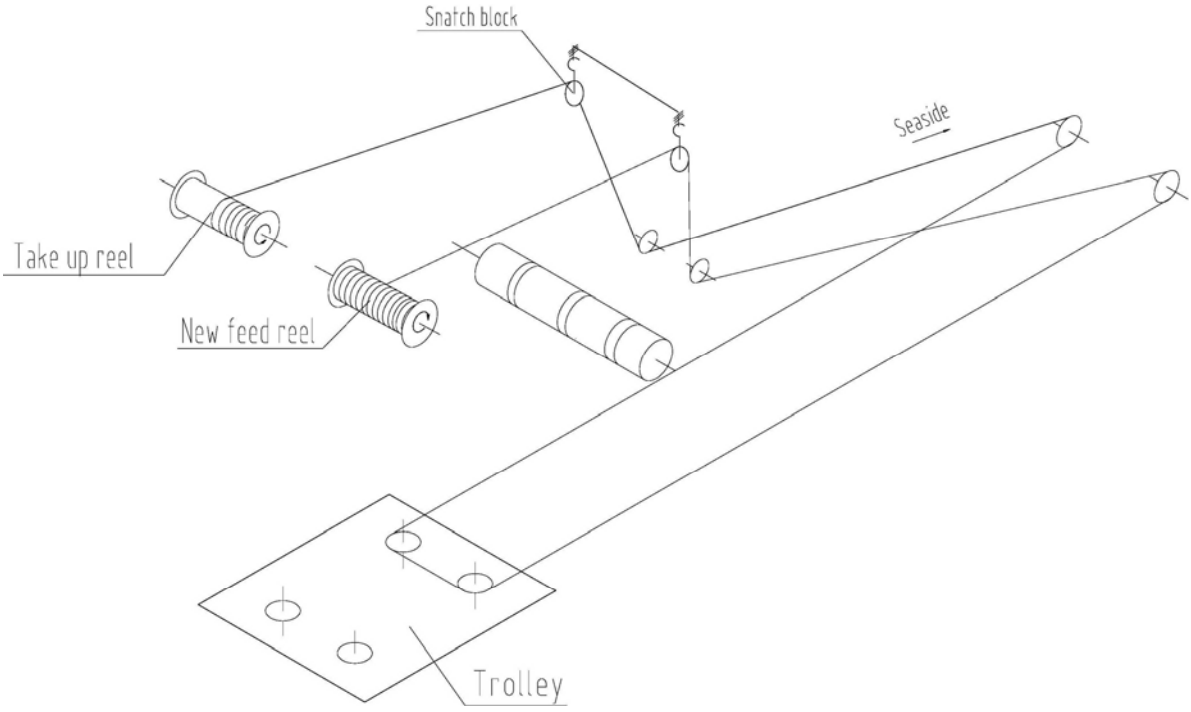


Figure 4-49:：Step 10

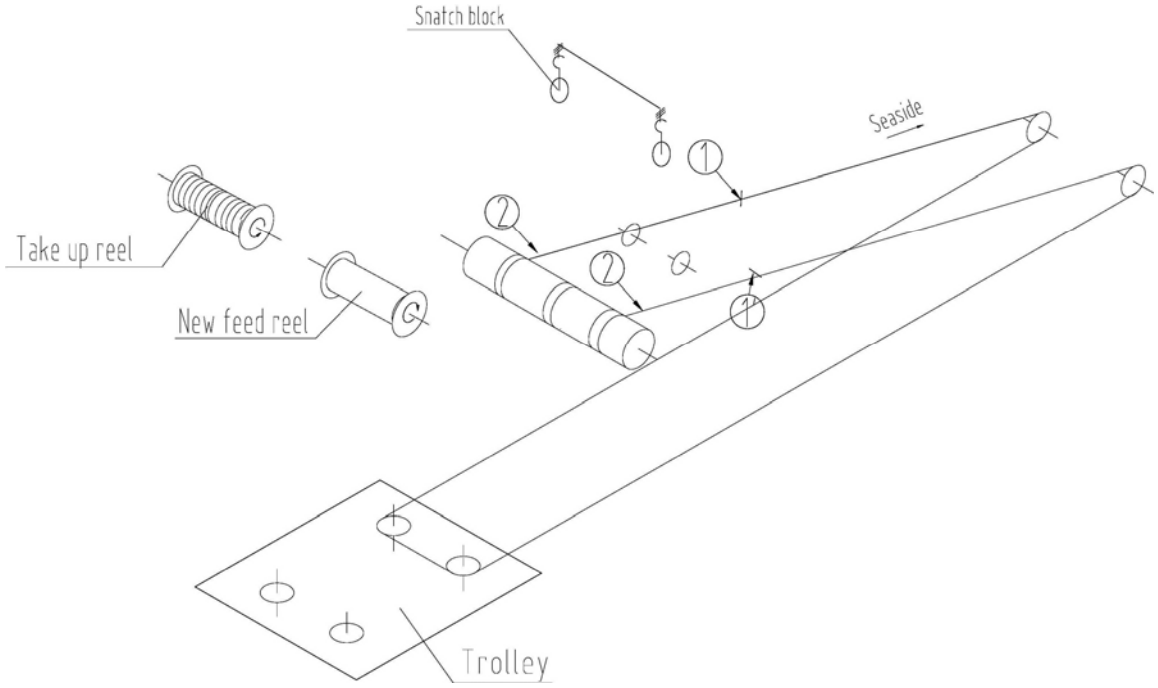


Figure 4-50: Step 11,12,13

4.5.4 Rope re-reeving of boom hoist rope

> **Since natural corrosion is the major cause of failure for boom hoist wire rope, ZPMC request that the boom hoist wire rope be replaced every eight years.**

1) Move the crane to a suitable position and lower the boom to level. Then hang a warning signs to warning ships if necessary.

2) Move the reels with new ropes into the machinery house. Arrange the snatch sheaves. See Figure 4-53.

3) Secure the wire rope at the output point (See Figure 4-533) of machinery house. The boom hoist wire rope are heavier so four 2.5t chain block shall be used. Make sure the rope clamp and the chain block can hold the whole rope safe.

4) Release the rope clamp at A-frame.

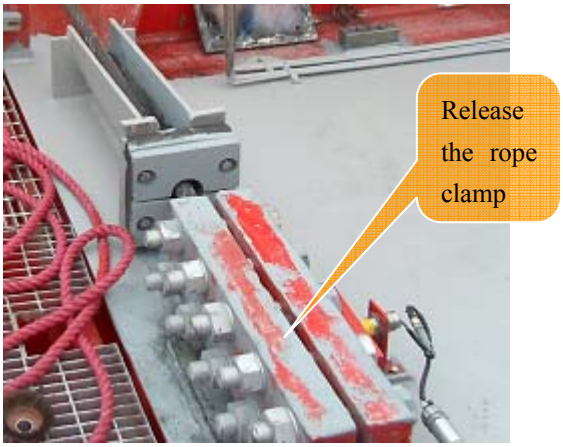


Figure 4-51: Rope clamp at top of A frame

5) Record the number of rope wraps on the drum “n” (shall not be less than 5). After taking the rope off from the drum ,measure and record the rope length “ L” inside the machinery house roughly

6) Pull the old rope to go through the guide sheave and connect the old rope end (driving reel side) with new ropes. And connect the old rope at driving side to the driving reel. Then run the reel and pull the old rope in slowly until the rope is slightly tighten and then hold

the reel by manual brake.

7) Slowly release the securing point outside of the machinery house. The whole system will be the same as Figure 4-544.

8) Run the reel and pull the old rope slowly until about new rope in "L" length go into the machinery house. Control the speed of reel by manual brake. There shall be two inspectors at the top of APEX and boom tip to observe if there are anything that may damage the rope occurs. If the reel is stop, hold the bind brake tightly and make sure there's no movements any more.

9) Secure the point outside of the machinery house again. Disconnect the new and old ropes and Install the driving reel side new rope to the drum about "n" Wraps. Install the rope clamps on drum and tight to 50% of the finally torque preliminary.

10) Release the security point outside of the machinery house.

11) Adjust the rope tension by chain block. Make sure the slack rope sensor on boom tip is working properly. The sensor device may be damaged if the rope are loosing excessively due to the heavier tare weight of rope. See



12) Figure **4-52**2 for reference.



Figure 4-52: rope slack sensor

13) Payout the remaining new rope on idle reel. and install the rope end to the drum and make sure there are at least 5 wraps on the drum and then secure it by rope clips and tighten to 50% of finally tightening torque

14) Release the rope security here. Check if the rope reeving system is OK. If not, adjust the rope tension by chain block again and switch the system to normal operating.

15) Check if the cam is working properly (Disconnect the cam before rope re-reeving.)

16) Boom up to 80 degree and see if there are 1~2 empty rope grooves on the drum. Check whether the rope slack sensor works properly. If not, the rope shall be adjusted: If the empty groove is less than one, estimate the rope length need to be adjusted and then boom down to level. Secure the wire rope by clamp and chain block near by drum. Then loose the rope clips on drum and pay some rope out then tight the rope clip again. Release the rope security of chain block. Similar procedure can be applied if the rope slack sensors are not work properly due to the incorrect rope slack.

17) Lubricate the new rope. This can also be done during the rope changing simultaneous but additional manpower may be needed.

Table 4-13: Bolt tighten torque

Location/size	Toque
Rope clamp at top of A frame /M30，10.9s	1150Nm
Rope clips on durm/M36	1450 Nm

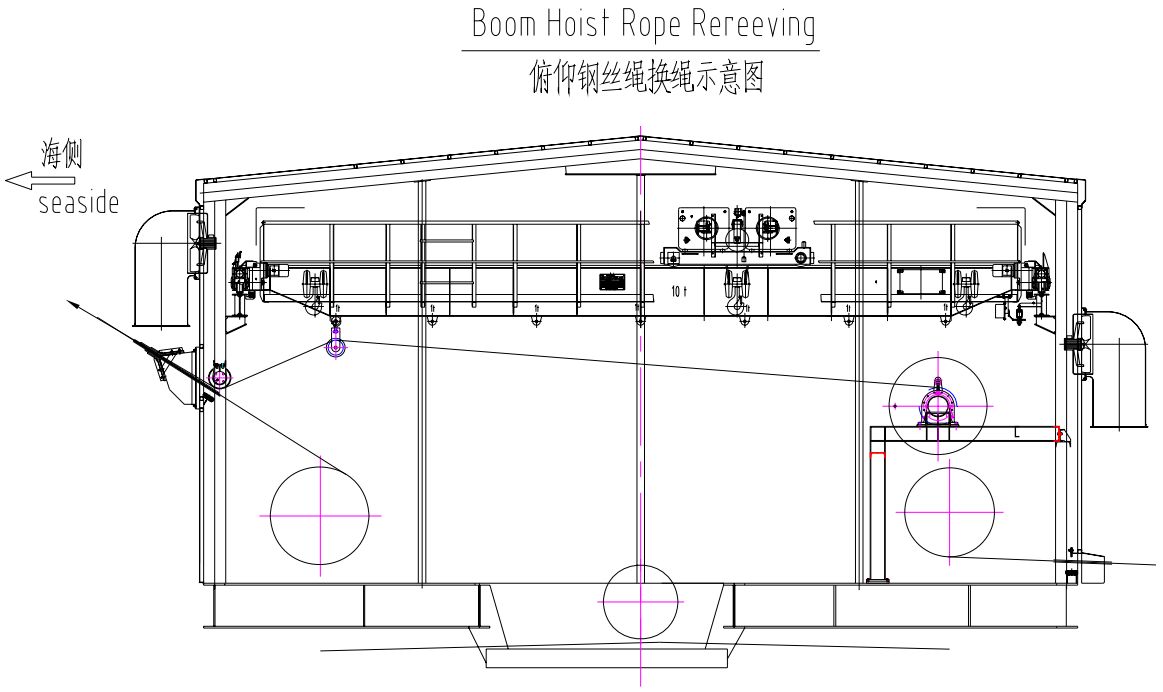


Figure 4-53: Snatch block arrangement inside machinery house

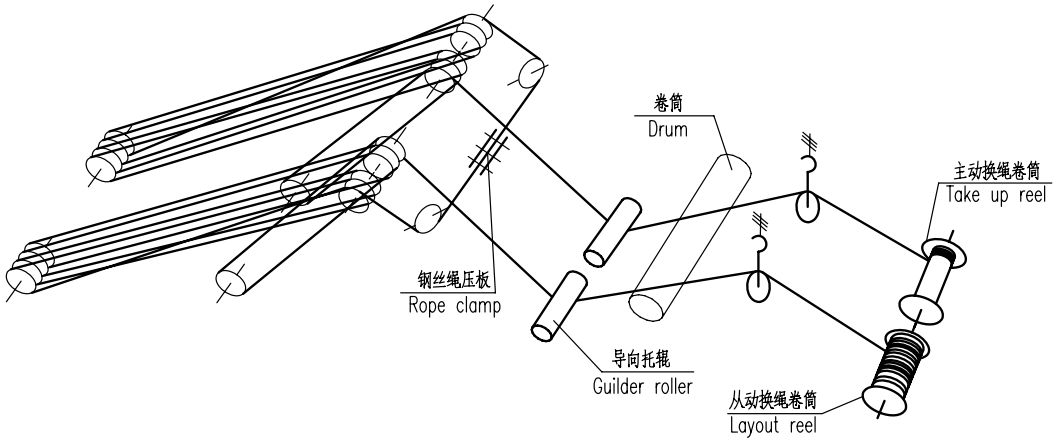


Figure 4-54: Schematic of Boom hoist re-reeving system

4.5.5 Rope re-reeving of Catenary trolley rope

Detail procedure of re-reeving of Catenary trolley towing rope are below. It is assumed that the rope which needs to be changed is at shorter. The procedures are similar if the other rope needs to be changed.

1) Move the main trolley to the park position. The catenary trolleys move along with the main trolley to a certain position. The boom shall be in level position. Switch the crane to maintenance mode.

2) Release the hydraulic pressure of trolley rope tension by opening break valve.

3) Move the reel with new ropes into the machinery house and reel it on the one side.

4) On the trolley, Loosen the rope clamps of the rope and take the rope end out of the clamp and on the roller.

5) Install two small sheave near the hinge cross beam which is connected with two side independently rope.

6) Released the rope end on the boom hinge, then two rope end are connected by weld and the rope turn on the sheave.

7) Installed the guide rollers as shown on figure 4-55

8) Towing the new rope end to the seaside catenary trolley ,Connect end by weld, same way, towing the old rope end from other side catenary trolley to the re-reeving drum.

9) Run the rope re-reeving device, reeve the used rope onto the steel drum via towing rope, while new rope is spooled out via towing rope and catenary trolley rope reeving system onto the catenary trolley drum.

10) After the new rope comes onto the catenary trolley, separate the connection between wire rope and towing rope.

11) Fix ends of the rope for all where.

12) Take away the guide roller from in front of main trolley drums to their normal position, that is in front of the boom hoist drum.

13) Adjust the length of the catenary trolley rope, make sure the catenary trolley wheels do not rub with the side of the rail.

14) Catenary trolley rope reeving is completed.

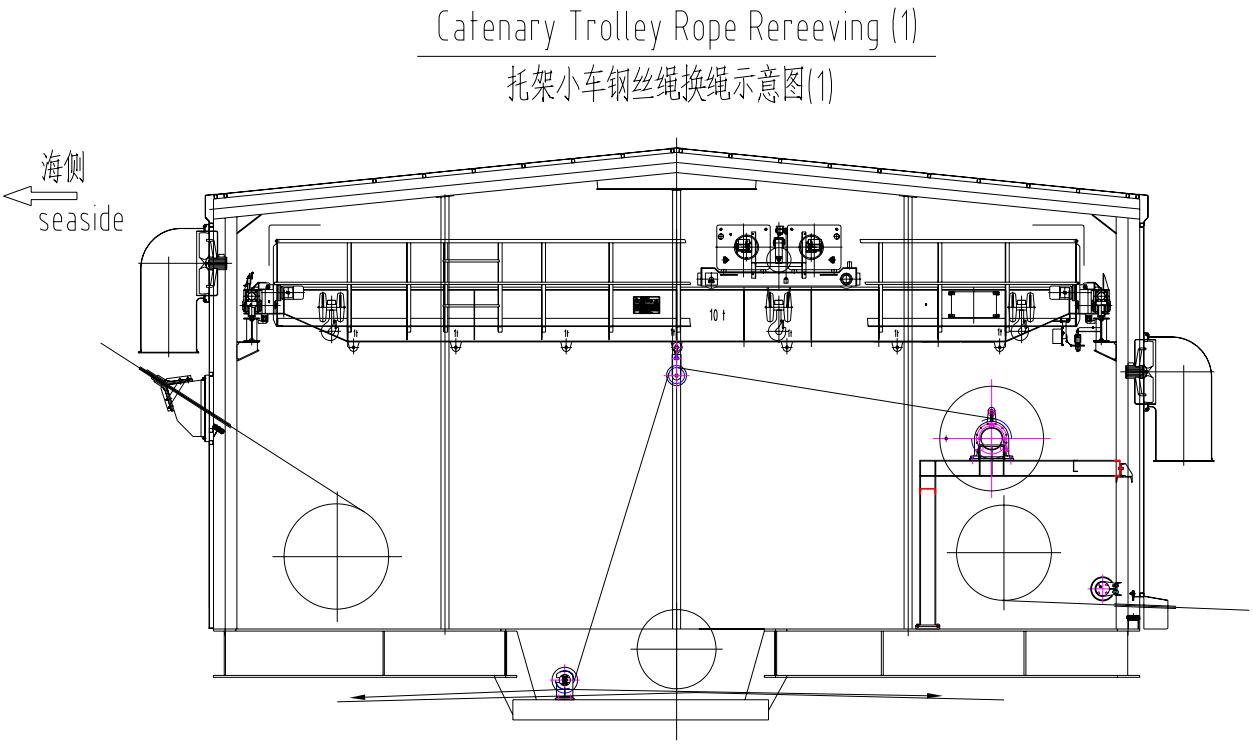


Figure 4-55: Snatch block arrangement inside machinery house

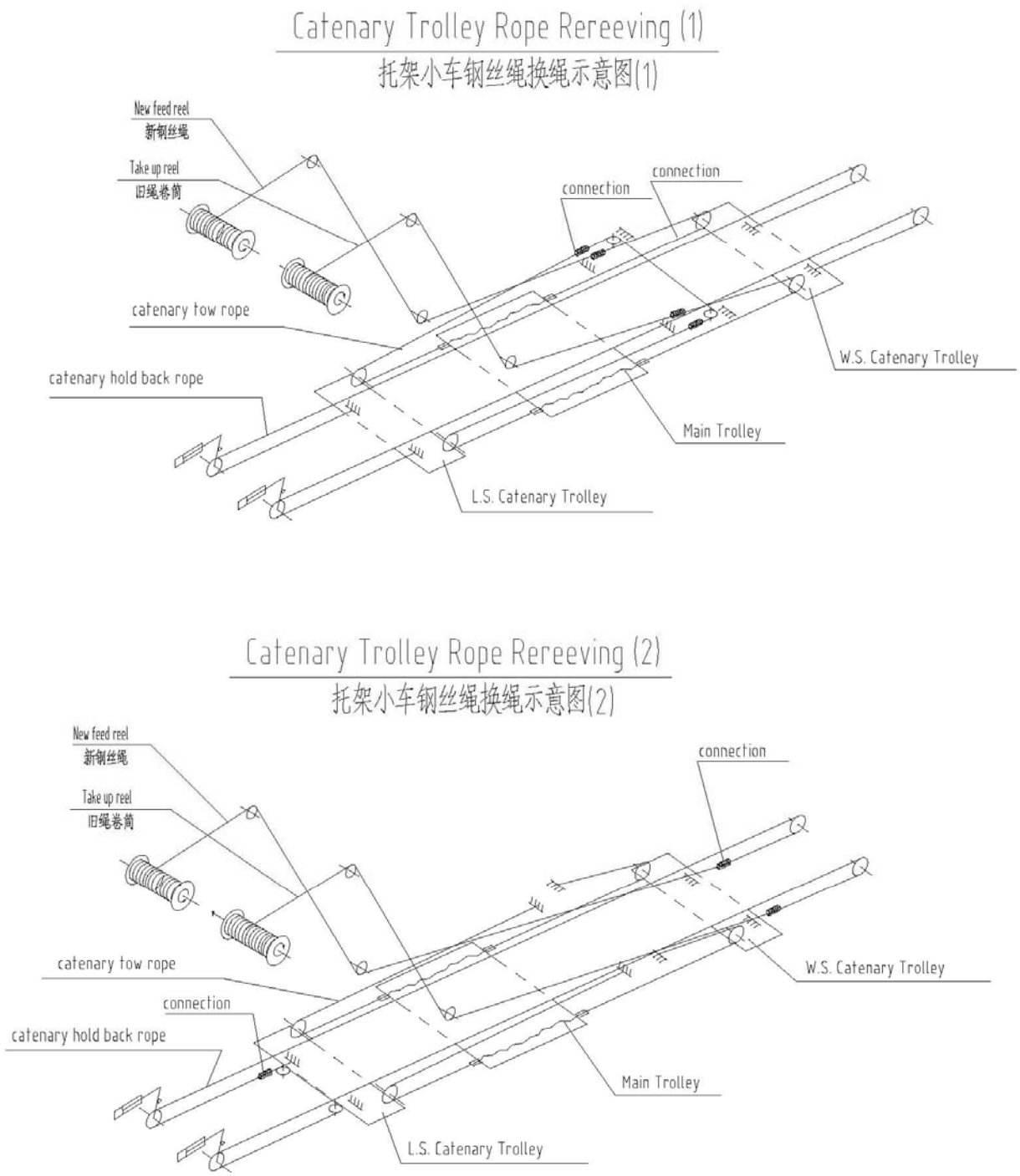


Figure 4-56: Schematic of Catenary trolley re-reeving system

4.6 Re-adjusting Rope Lengths



Danger of falling!



Danger of tripping!

The rope will be extended sometime after operating. Usually the rope need to be adjusted in case of the wire rope extend more than 150mm or the rope tension arm is detected by the sensor due to the rope extension, or the sensor at the top of A frame is activated, or when there are now empty rope grooves on boom hoist drum when boom fully up and etc. The adjustment of boom hoist rope has been instructed in section 4.5.4 step 15 and will not be included hereafter. Table 4-14: Manpower and tools for the rope adjustment

Table 4-14: Manpower and tools for the rope adjustment

Description		Quantity	Notes
Manpower	Mechanical	2	With radios and one act as supervisor
tools	Duster cloth	Some	For clean
	Torque wrench	1	
	monkey wrench	1	Match M20, M24, M36
	Special rope clamp	5	
	3t chain block	2	
	Wires, steel plate	some	Secure the rubber, wood plate, slap block
	Chalk/ Punched tape	1	For marking
	Note book/pen		For record

4.6.1 Re-adjusting the Main Hoist Rope Lengths

1) Move the trolley to the parking position.

2) Put the spreader on the ground until slack-rope limit switch stops the drive.

3) Lock out the crane's main hoist motion by carrying out the lock-out and tag-out procedure.

4) Secure the rope adjacent to the drum by using an auxiliary rope and chain hoist to unload the rope on the drum.

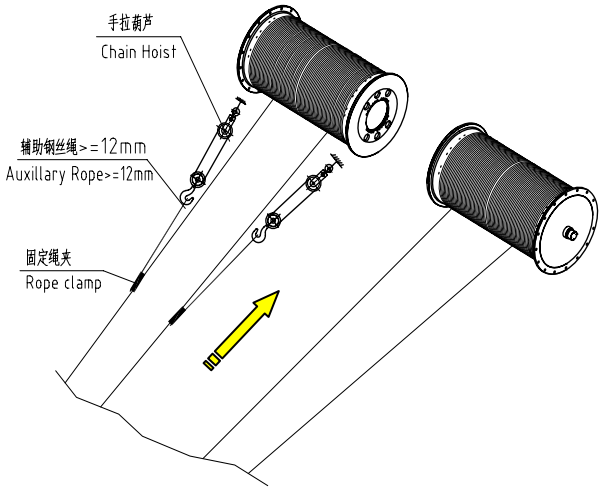


Figure 4-57: Step 4

5) Mark rope position adjacent to the clamp and remove rope clamps.

6) Slip the rope on the drum by appropriately the amount needed to square and level the spreader.

7) Re-install rope clamps and tightened to the correct torque. See Table 4-11 for specific torque value.

8) Repeat this procedure for all ropes as necessary.

9) Tie off the part of the rope which projects over the clamp with stainless wire and cut it off behind the tied off spot.

10) Remove all tools and lift equipment.

11) Return the crane to service after verifying the adjustment.

4.6.2 Re-adjusting the Trolley Towing Rope Lengths

	**No wheel flange riding on rail is permitted! Eliminate flange riding right away if it exists**
	**To avoid trolley rope breakage, always reserve the proper tensioner cylinder stroke to compensate the rope extension when boom is raised from level to stowing or maintenance position!**

The new trolley towing rope will be somewhat stretched after its initial 50 hours of vessel operation. The stretch extent is different from each trolley towing rope. If only a single rope is changed or if the two ropes stretch different amounts, the length difference of the waterside rope will cause the trolley not to run square with the rails. This will cause abnormal noise, and wheel flange and rail wear. The waterside ropes must be maintained to keep the trolley running square. The tolerance in the landside ropes is much more forgiving because of the tensioner. The landside ropes do not steer the trolley. They must be maintained within the travel length of the tensioner.

Wheel flange riding on the rail will damage the wheel, rail and the driving system. Flange interference will increase the travel resistance force up to as much as 3 times of the normal operation and will overload its driving system. Severe flange riding will make wheel flange runs to top of the rail and may cause trolley derail.

The position of each tensioner cylinder is monitored by two proximity switches. The original position of the cylinder is with the sheave arm almost vertical under the pressure of hydraulic system. The arm may offset from its original position and trolley motion is allowed until either of the limit switches is activated.

If either of the limit switches is activated, the landside trolley rope length adjustment is needed. Uneven waterside rope lengths cause trolley wheel flanges to ride on the rail, the waterside relative rope length can be measured as per the steps mentioned in Section 4.6.3. Adjust the waterside trolley rope length accordingly.

Steps to adjust the trolley rope length are as follows:

1) If a waterside rope is to be adjusted, move the trolley to the backreach to remove as much of the rope from the drum as possible. If a landside rope is to be adjusted, move the trolley to the full outreach.

2) Release the trolley rope tensioner at the backreach. Fully open this valve so that the rope tensioner cylinders will be released. See Section 5 for detail.

3) Lockout the trolley drive and the multifunction power unit starter by performing the lockout/tag out procedure.

4) Secure the wire rope which needs adjustment in the machinery house by holding the rope leading from the drum using an auxiliary rope and chain hoist. See Figure 4-57.

5) Mark the rope position adjacent to the clamp. Loosen the wire rope clamps on the drum for the rope which needs adjustment.

6) Slip the appropriate length of wire rope out of the drum.

7) Tighten the rope clamps onto the drum.

8) Wrap the rope end with stainless steel wire at 1in. away from the cutting point.

9) Cut off the extra length wire rope.

10) Fully close the relief valve which was

mentioned in Step 2.

11) Turn on the control power to watch the position of the tensioner arm.

12) Repeat the same procedure for the other rope ends if necessary.

13) Remove all tools and lift equipment.

14) Run the trolley over the majority of its travel length several times while monitoring the trolley tracking, the tensioner, and the tow rope position.

15) Return the crane to service after verifying the adjustment.

4.6.3 Method for Measuring Relative Waterside Trolley Rope Length

Steps for measuring relative waterside trolley rope length are as follows:

1) Run the trolley to the boom tip so that the waterside trolley rope can be wrapped into the drum as much as possible.

2) Inspect the trolley wheel to rail to ensure the trolley is square to the rail. If necessary re-position the trolley to make it square

3) Release the trolley rope tensioner at the backreach. Fully open this valve so that the rope tensioner cylinders will be released. See section 5 for details.

4) Lockout the trolley drive and the multifunction power unit starter by performing the lockout tag out procedure.

5) Remove the sheave rope guards at boom tip.

6) Secure and hold the waterside trolley rope leading from the drum using an auxiliary rope and chain hoist at boom tip as shown in Figure 4-58.

7) Tighten the two waterside trolley ropes with the same force produced by both chain hoists.

8) Inspect the two trolley rope catenaries leading from the trolley to the boom tip. And measure the two trolley rope length difference.

9) Remove all tools and equipment.

10) Adjust the waterside trolley rope length as necessary according to the procedure mentioned in Section 4.6.2.

11) Remove lockout/tag out devices and operate the trolley over the majority of its travel distance several times. During each movement monitor the trolley squareness with the rails. Rope adjustment may need to be repeated if the trolley fails to run square.

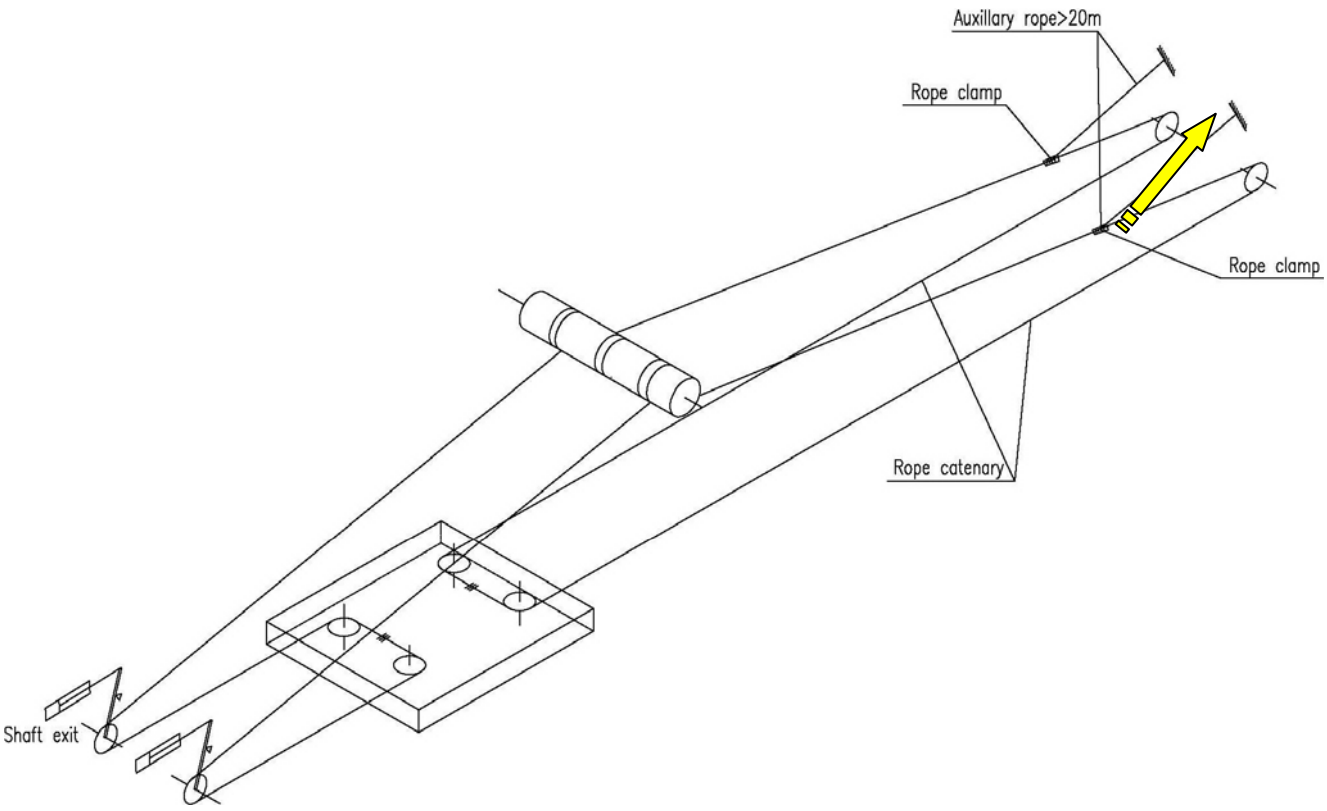


Figure 4-58: Measurement of rope difference between landside and waterside

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